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Module 1: Overview of System Center Configuration Manager (SCCM)

What is SCCM?

SCCM (System Center Configuration Manager) – now called **Microsoft Endpoint Configuration Manager** – is a tool developed by Microsoft that helps IT administrators manage computers, servers, and devices across an organization from a **central location**.



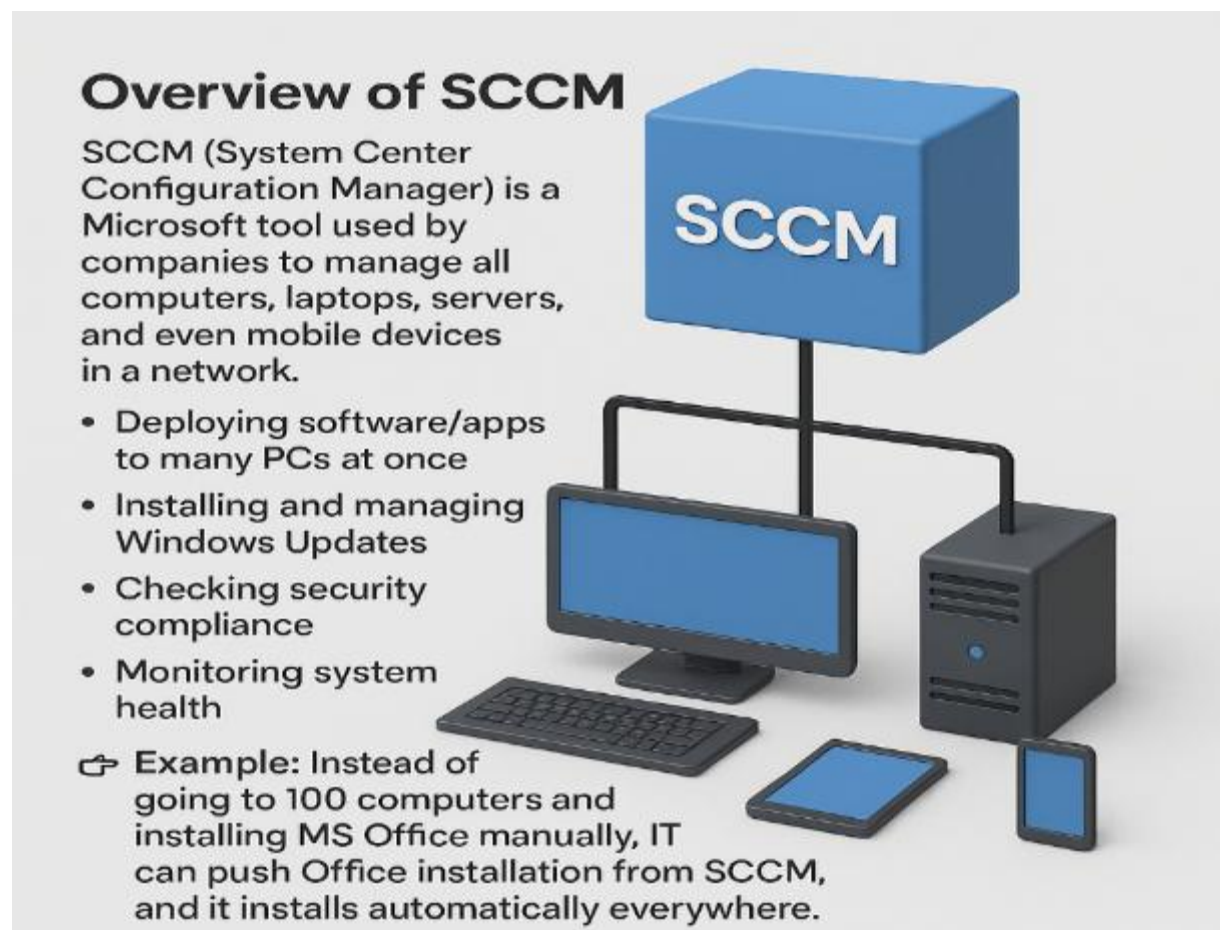
1. Overview of SCCM

SCCM (System Center Configuration Manager) is a Microsoft tool used by companies to manage all computers, laptops, servers, and even mobile devices in

a network.
It helps IT admins in:

- Deploying software/apps to many PCs at once
- Installing and managing **Windows Updates**
- Checking security compliance
- Monitoring system health

👉 Example: Instead of going to 100 computers and installing MS Office manually, IT can push Office installation from SCCM, and it installs automatically everywhere.



2. Benefits of Implementing SCCM

- ✓ Centralized management of all devices
- ✓ Saves time and reduces manual work
- ✓ Ensures all systems get updates at the same time
- ✓ Improves security (patching vulnerabilities quickly)
- ✓ Generates reports (like which computers need updates)

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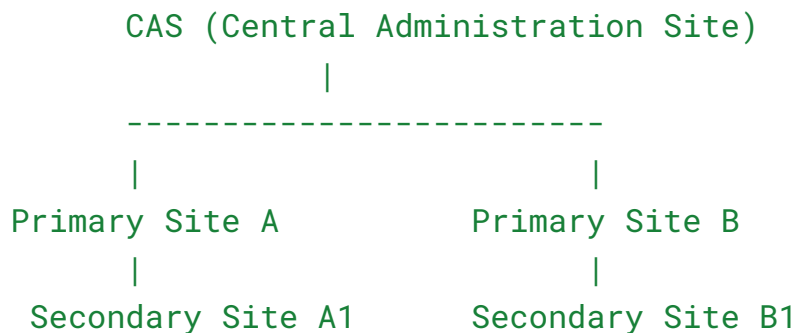


3. Hierarchy in SCCM

Hierarchy = Structure of SCCM environment.
It can have **1 or many sites**, depending on company size.

- **Central Administration Site (CAS)** → Used in very large organizations. Manages multiple primary sites.
- **Primary Site** → Main site that manages devices directly.
- **Secondary Site** → Optional site, used to manage remote offices with slow network connections.

Hierarchy Diagram (simple):

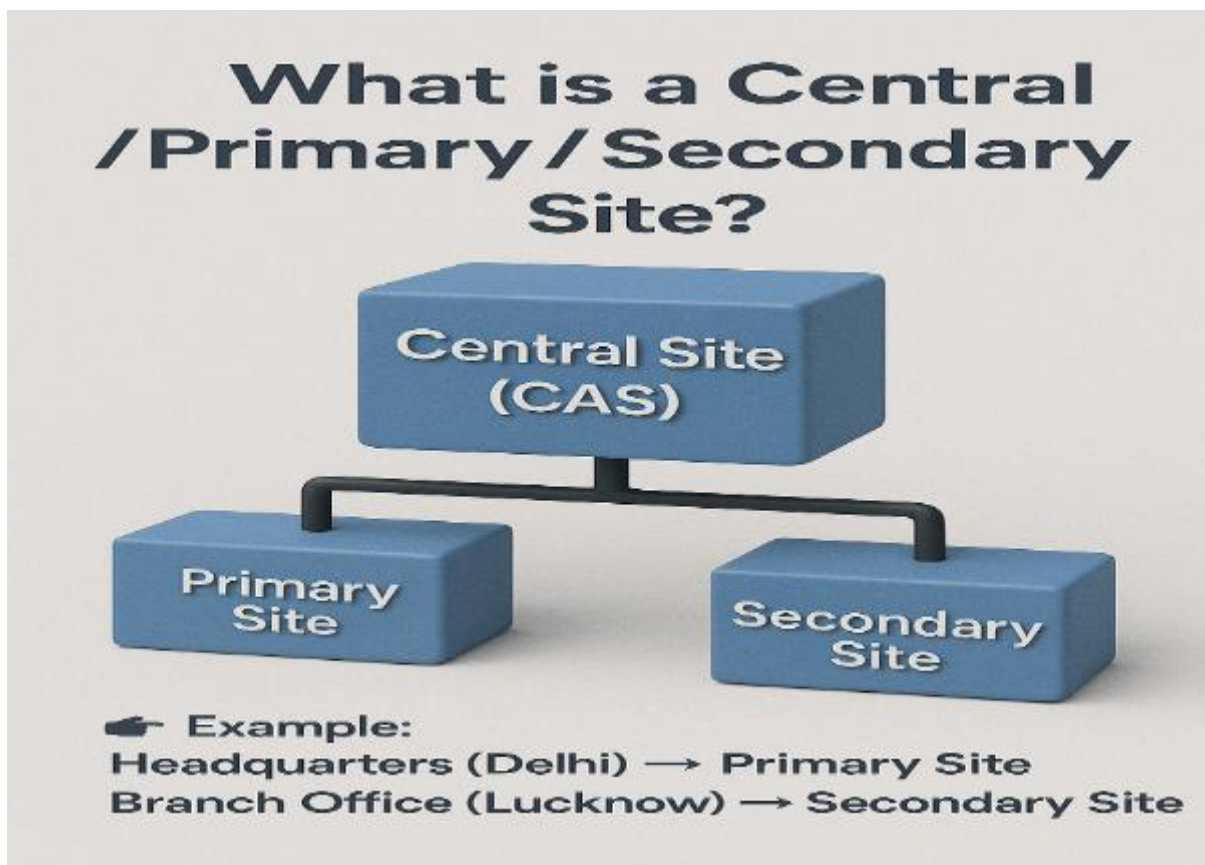


4. What is a Central / Primary / Secondary Site?

- **Central Site (CAS):** Top-level site; manages multiple primary sites.
- **Primary Site:** Can manage devices directly. Usually **1 per company** if not very large.
- **Secondary Site:** Supports a primary site; used in branch offices. Cannot exist without a primary site.

👉 Example:

- Headquarters (Delhi) → **Primary Site**
- Branch Office (Lucknow) → **Secondary Site**



5. Configuration Manager Site System Roles

Each SCCM site has **roles** (functions).

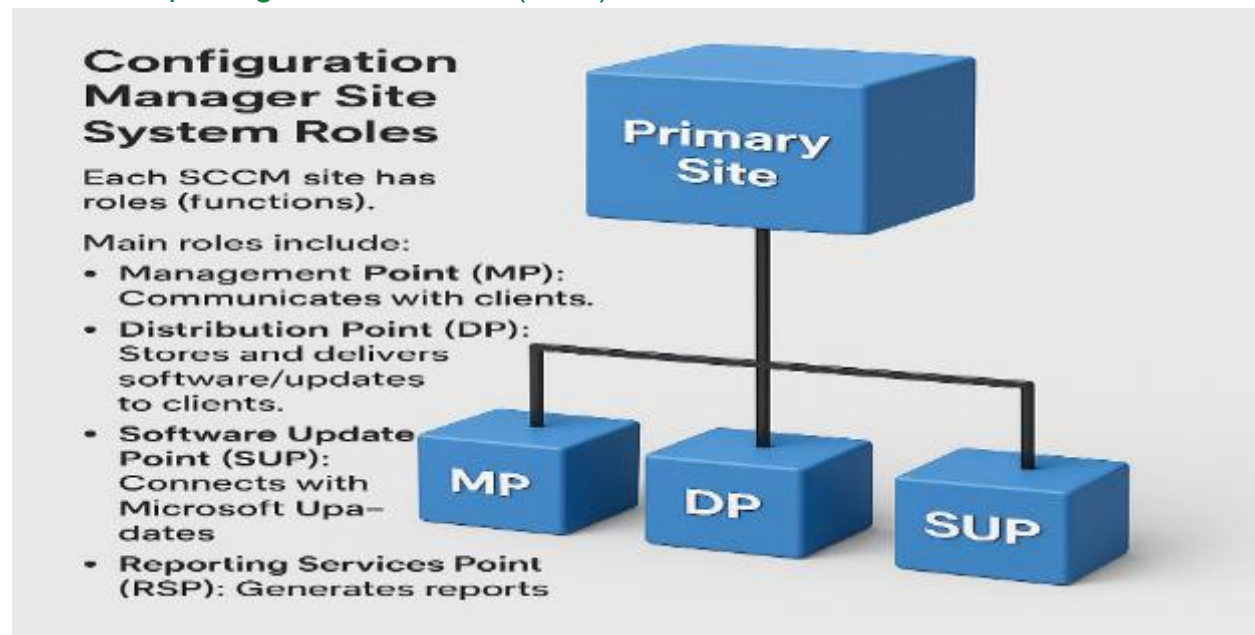
Main roles include:

- **Management Point (MP):** Communicates with clients.
- **Distribution Point (DP):** Stores and delivers software/updates to clients.
- **Software Update Point (SUP):** Connects with Microsoft Updates.
- **Reporting Services Point (RSP):** Generates reports.

Site Roles Diagram:

Primary Site

- Management Point (MP)
- Distribution Point (DP)
- Software Update Point (SUP)
- Reporting Services Point (RSP)

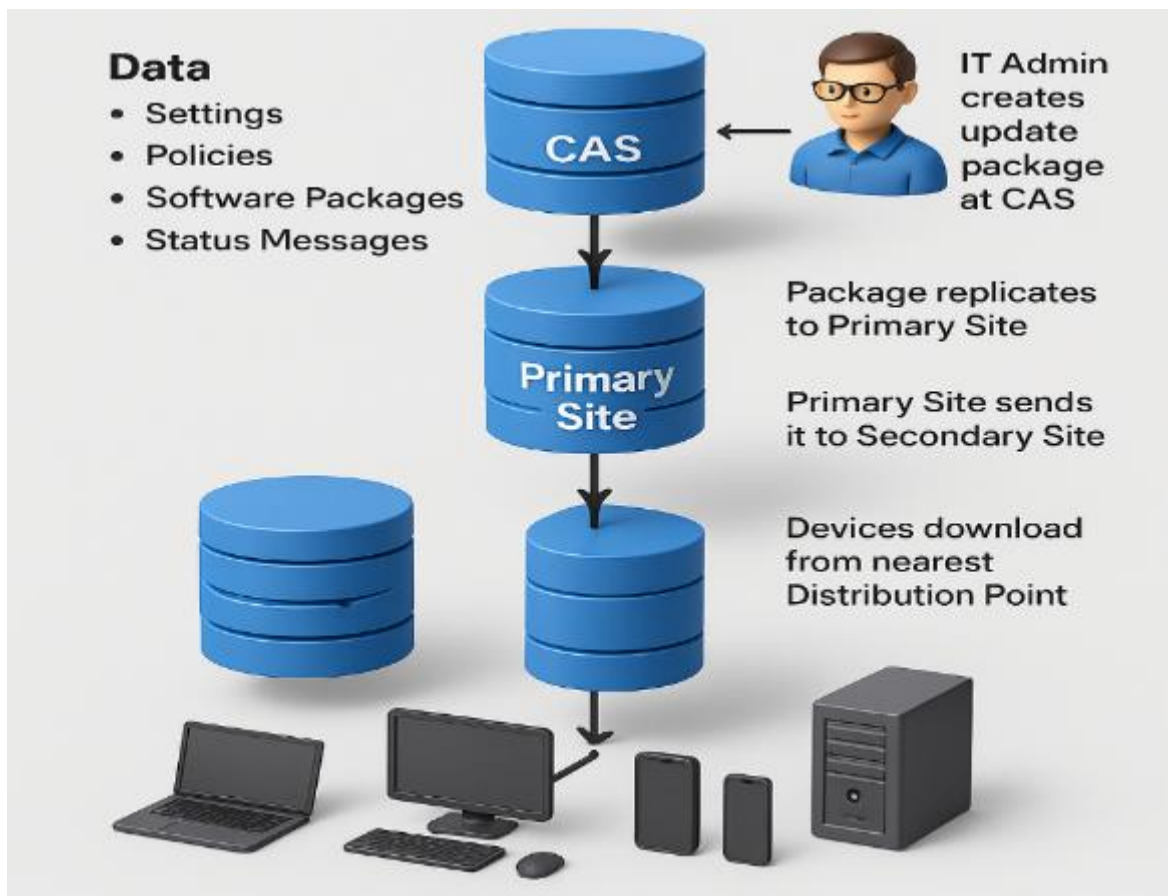


6. How Data Flows and Replicates in a Hierarchy

- Data = Settings, Policies, Software Packages, Status Messages.
- Data moves **up** and **down** the hierarchy.

🔑 Example of flow:

1. IT Admin creates update package at CAS.
2. Package replicates to Primary Site.
3. Primary Site sends it to Secondary Site.
4. Devices download from nearest **Distribution Point**.

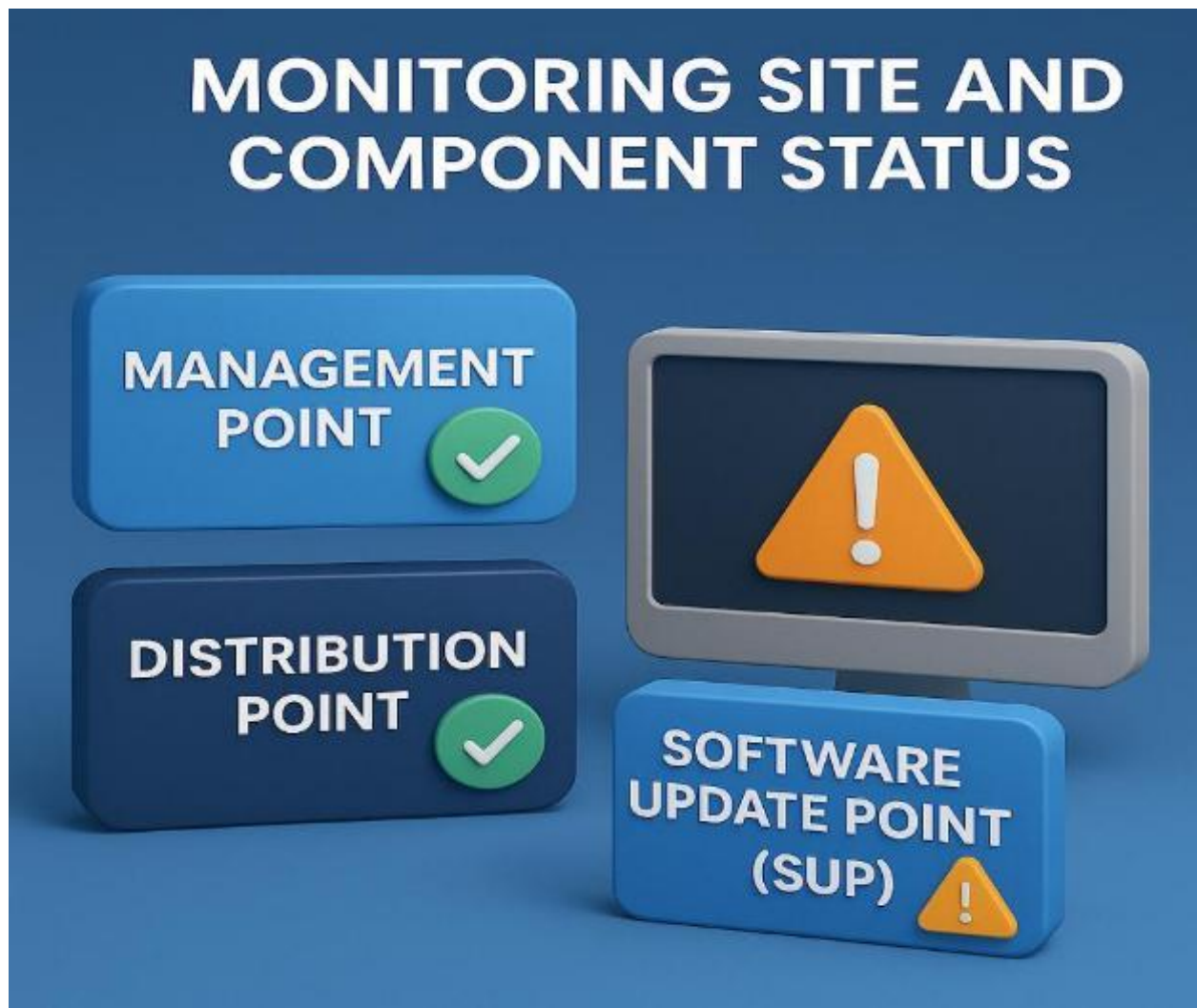


7. Monitoring Site and Component Status

SCCM includes built-in **monitoring tools** to check:

- Site health (working or not)
- Component status (like DP, MP, SUP)
- Alerts and logs

👉 **Example:** If the Distribution Point is not working, admins get alerts in SCCM console.



8. Planning an SCCM Stand-Alone Primary Site

Most companies don't need a **CAS**.

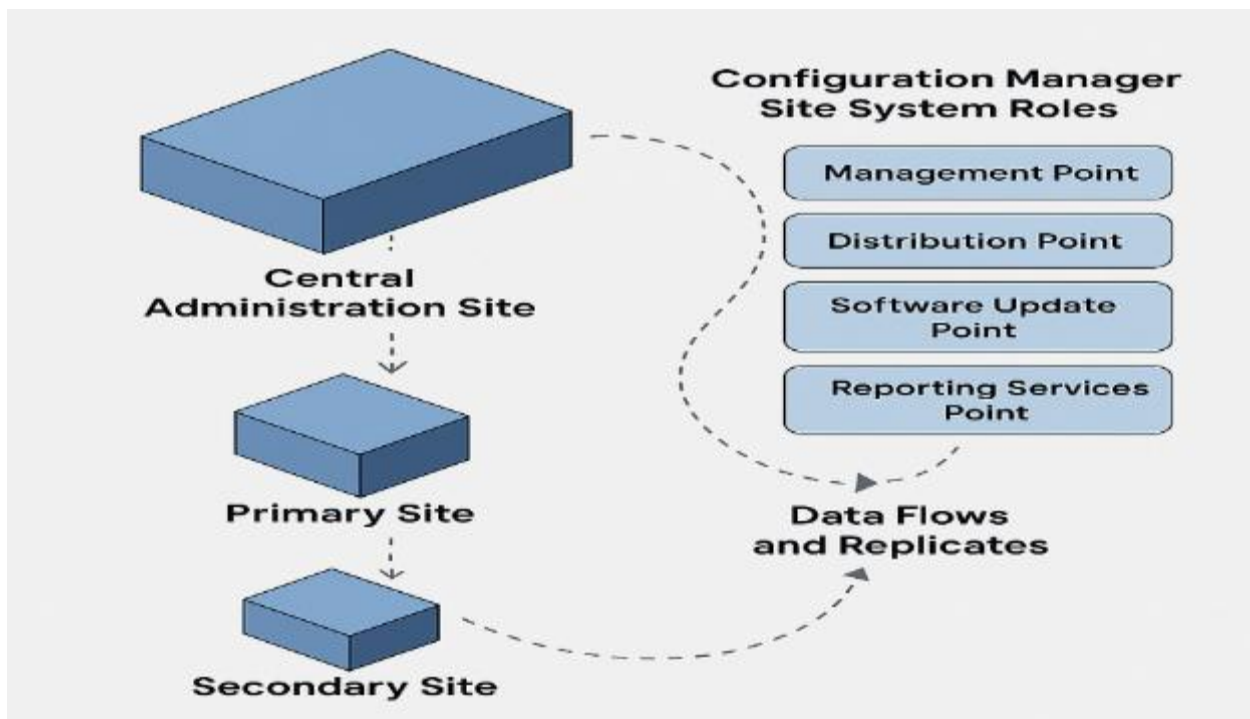
👉 A **Stand-Alone Primary Site** is enough for small–medium businesses.

Planning includes:

- **Server requirements** (CPU, RAM, storage)
- **Network bandwidth**
- **Deciding roles** (DP, MP, SUP)
- **Database setup** (SQL Server)

Example:

- Company with **2,000 computers** → Use only **1 Primary Site** with required roles.



Module 2: System Center Configuration Manager (SCCM) Primary Site Deployment

1. Exploring the SCCM Console with All Prerequisites

Before using SCCM, ensure the prerequisites are ready:

- **Windows Server** installed (2019/2022 recommended)
- **Active Directory (AD)** configured
- **SQL Server** installed
- **.NET Framework, IIS, and WSUS** installed
- **Firewall rules** configured

👉 After setup, launch the **SCCM Console**. This is the main admin tool where you manage clients, updates, apps, and reports.



2. Extending the Active Directory Schema

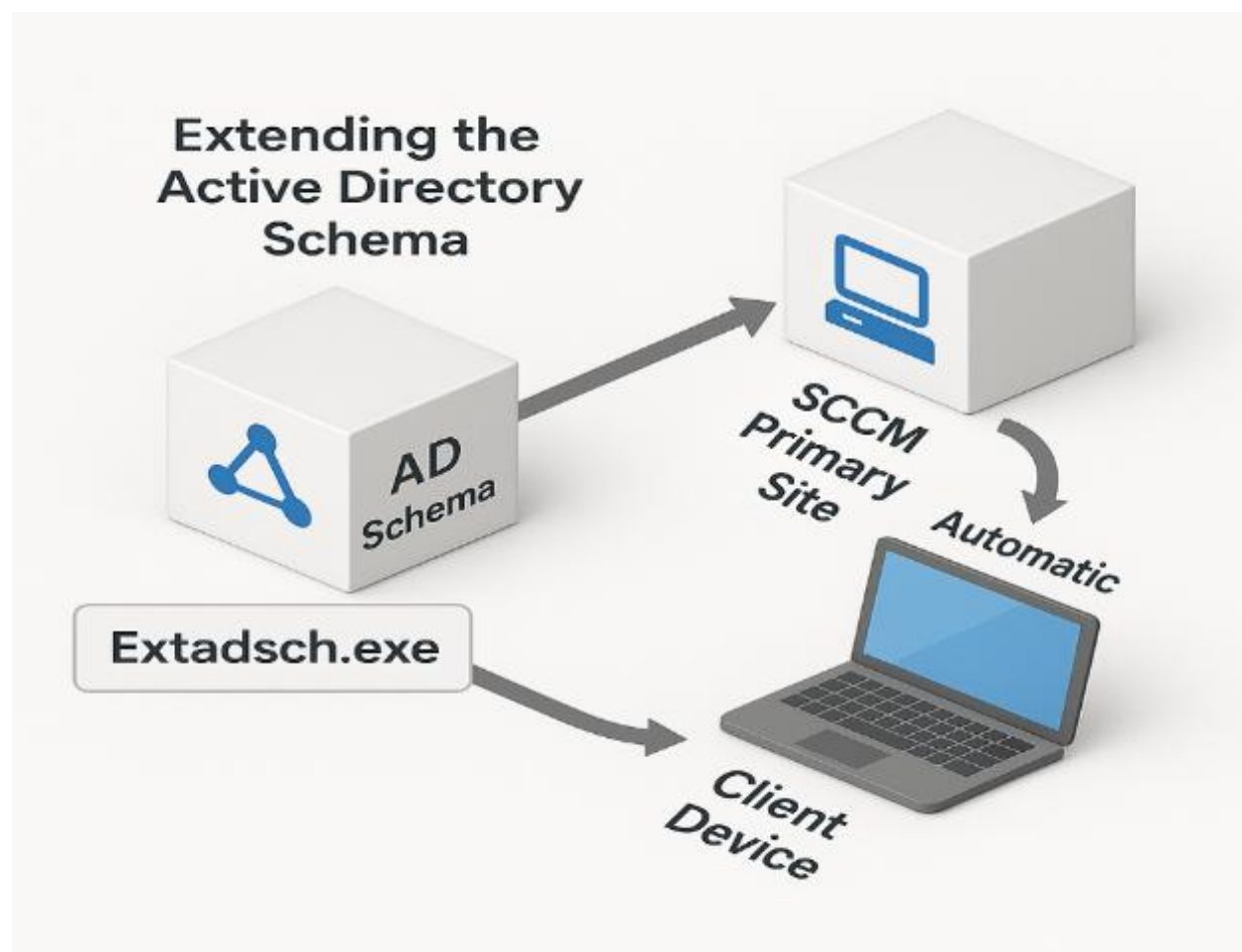
- Extending the **AD Schema** allows SCCM to publish site information into AD.
- Clients can automatically find their SCCM site from AD.

👉 Example: When a new laptop joins the domain, it can auto-discover the SCCM Primary Site.

Command used:

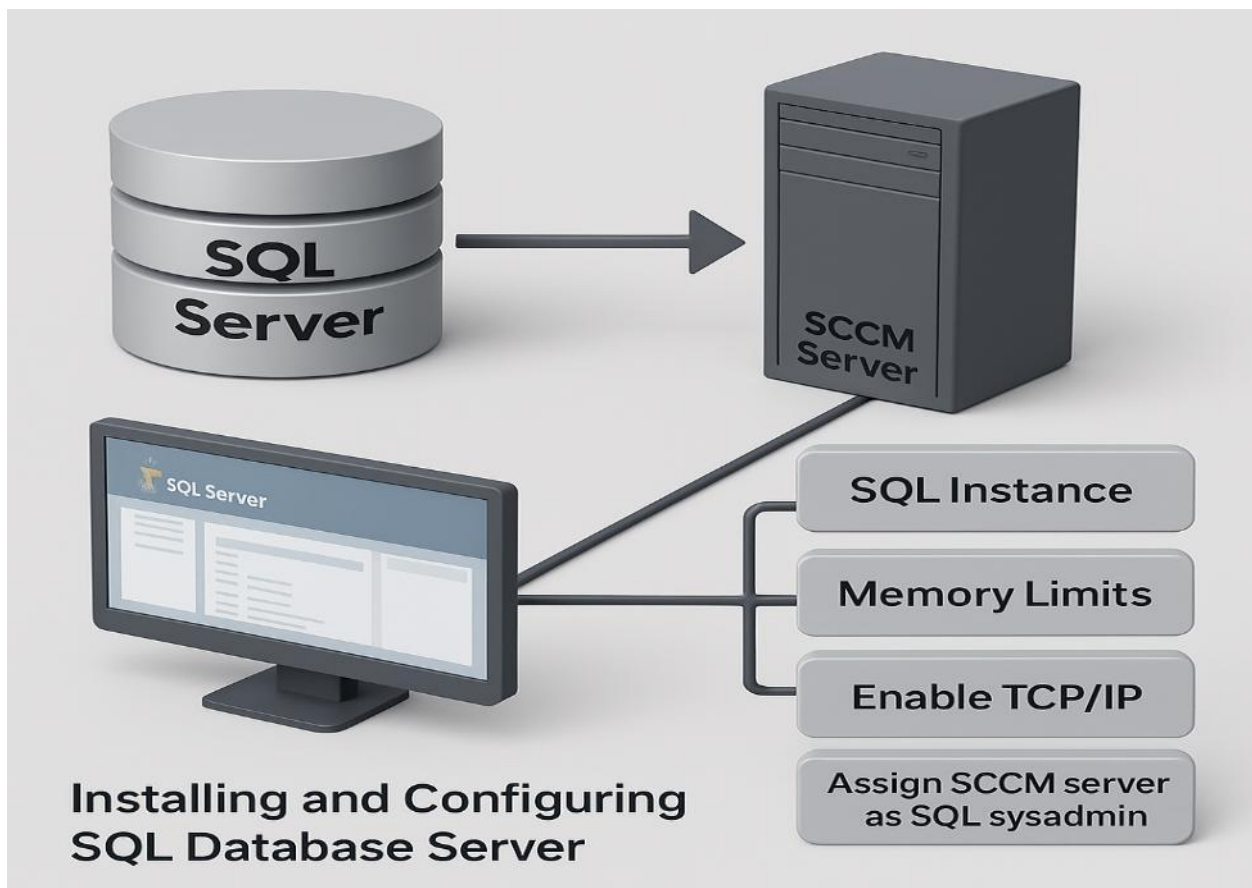
`Extadsch.exe`

(Log file generated: `extadsch.log`)



3. Installing and Configuring SQL Database Server

- SCCM uses **SQL Server** to store all its data (clients, updates, policies, inventory).
- Install SQL Server (on same machine or separate dedicated server).
- Configure:
 - SQL Instance
 - Memory limits
 - Enable TCP/IP protocol
 - Assign SCCM server as **SQL sysadmin**



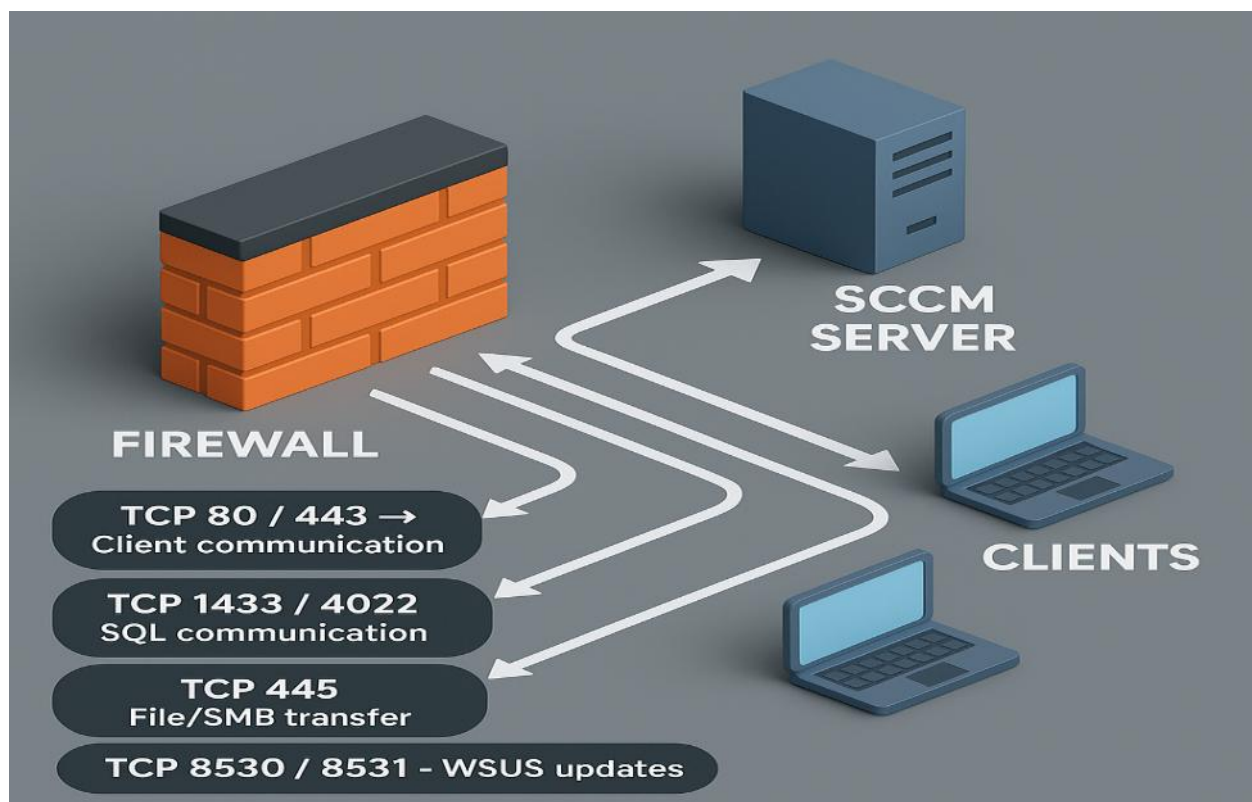
4. Configuring Network Firewall Settings for SCCM

1. **Purpose** → SCCM aur clients ke beech communication ke liye correct ports open hone chahiye.

2. **Important**

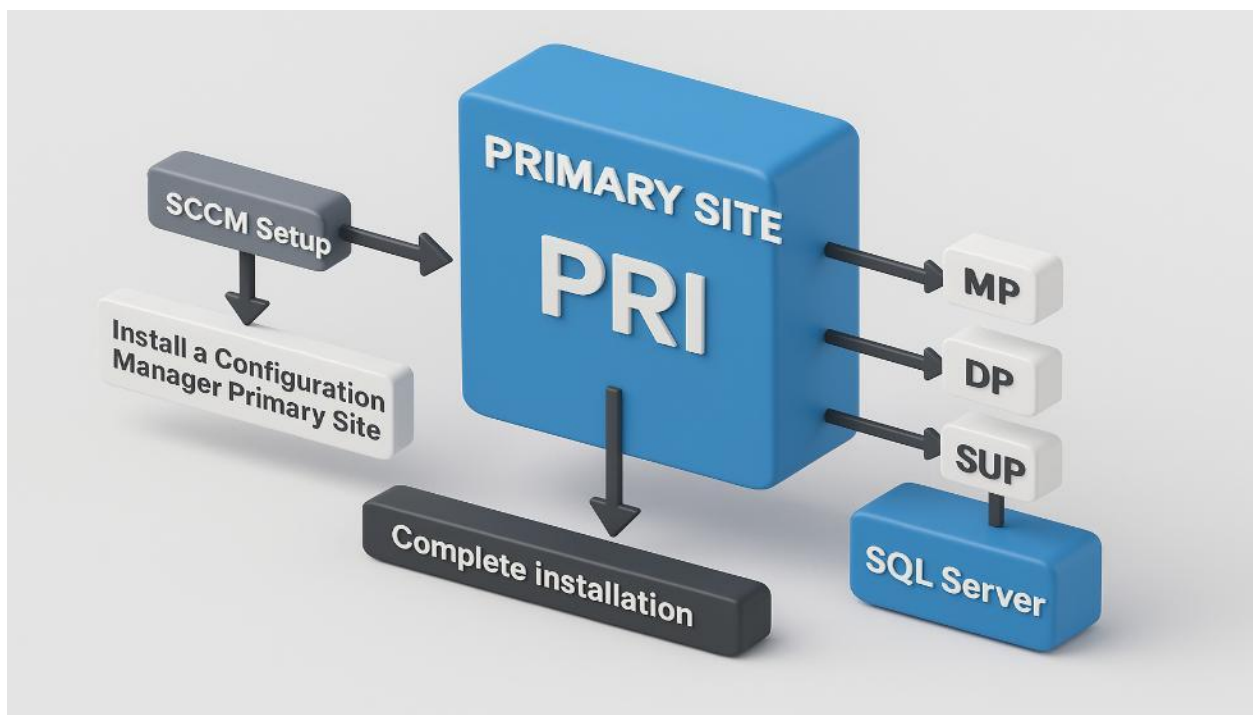
Ports:

- **TCP 80 / 443** → Client communication (HTTP/HTTPS)
 - **TCP 1433 / 4022** → SQL communication
 - **TCP 445** → File sharing / SMB transfer
 - **TCP 8530 / 8531** → WSUS updates
3. **Risk** → Agar firewall rules configure nahi kiye → Clients SCCM se connect nahi kar paayenge.



5. Installing an SCCM Site Server (Primary Site)

1. Run SCCM Setup
2. Choose Installation → *Install a Configuration Manager Primary Site*
3. Provide Site Code → 3 letters (e.g., PRI)
4. Define Site Name → e.g., HQ-Primary
5. Configure Site Roles →
 - MP (Management Point)
 - DP (Distribution Point)
 - SUP (Software Update Point)
6. Connect to SQL Server Instance
7. Complete Installation & Verify Logs

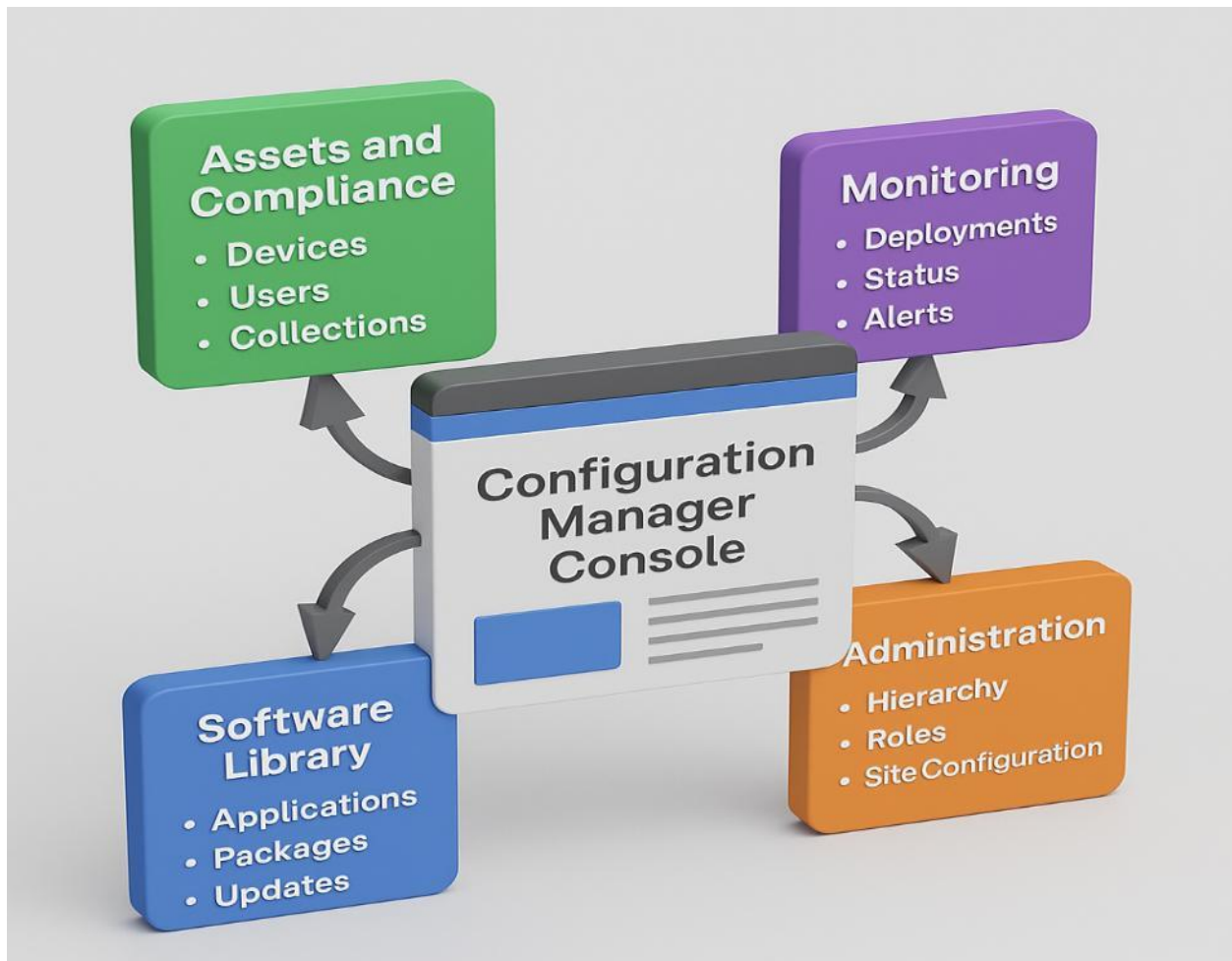


6. Exploring the Configuration Manager Console

The SCCM Console has **4 main workspaces**:

1. **Assets and Compliance** → Devices, Users, Collections
2. **Software Library** → Applications, Packages, Updates
3. **Monitoring** → Deployments, Status, Alerts
4. **Administration** → Hierarchy, Roles, Site Configuration

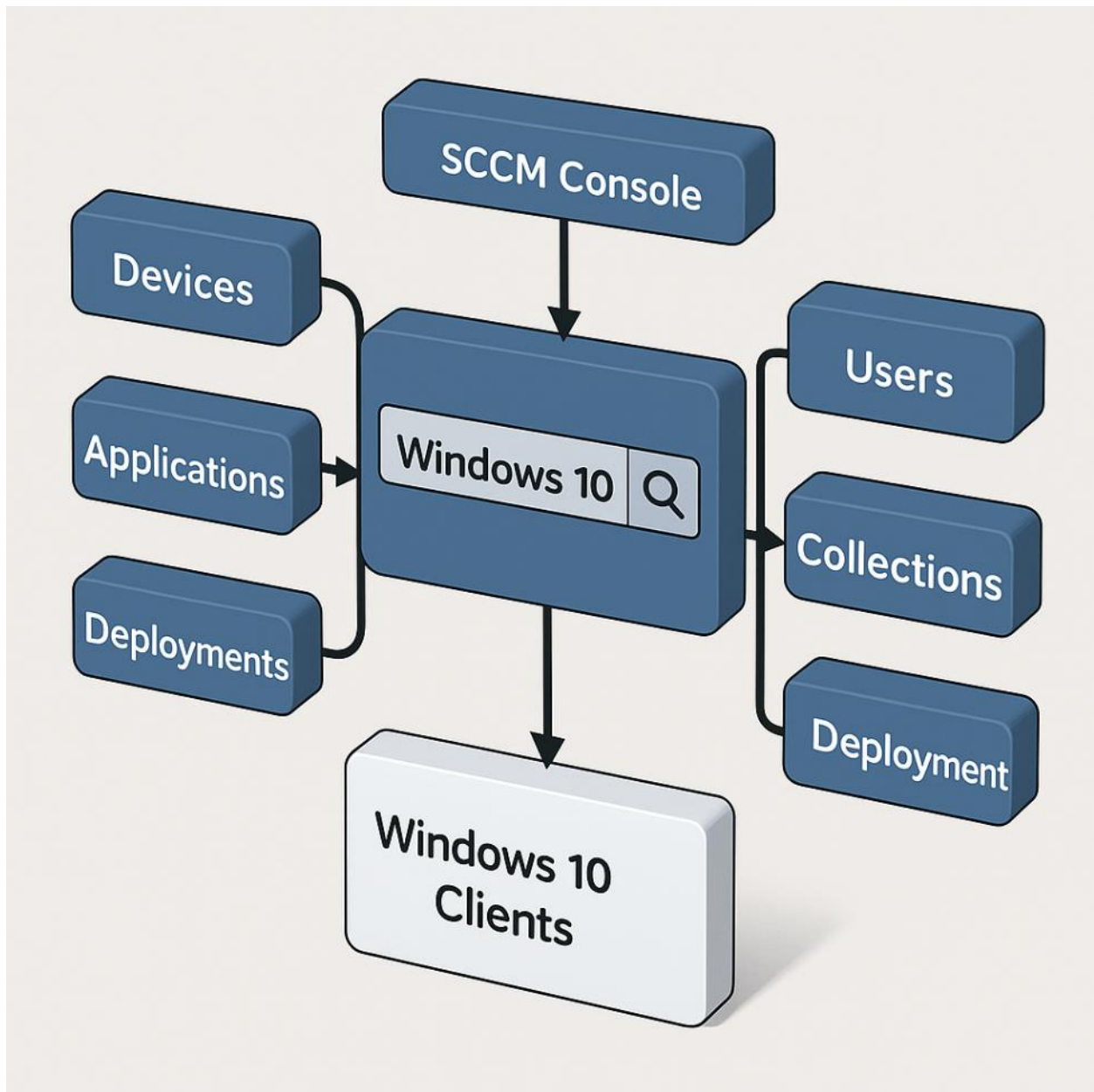
👉 This is where you do **90% of SCCM work**.



7. Searching in the SCCM Console

- You can search for **devices, users, applications, collections, deployments.**
- Supports filters (e.g., “All devices missing Office 365”).

👉 Example: Search → “Windows 10” → Lists all Win10 clients.



8. Monitoring and Troubleshooting a Configuration Manager Site

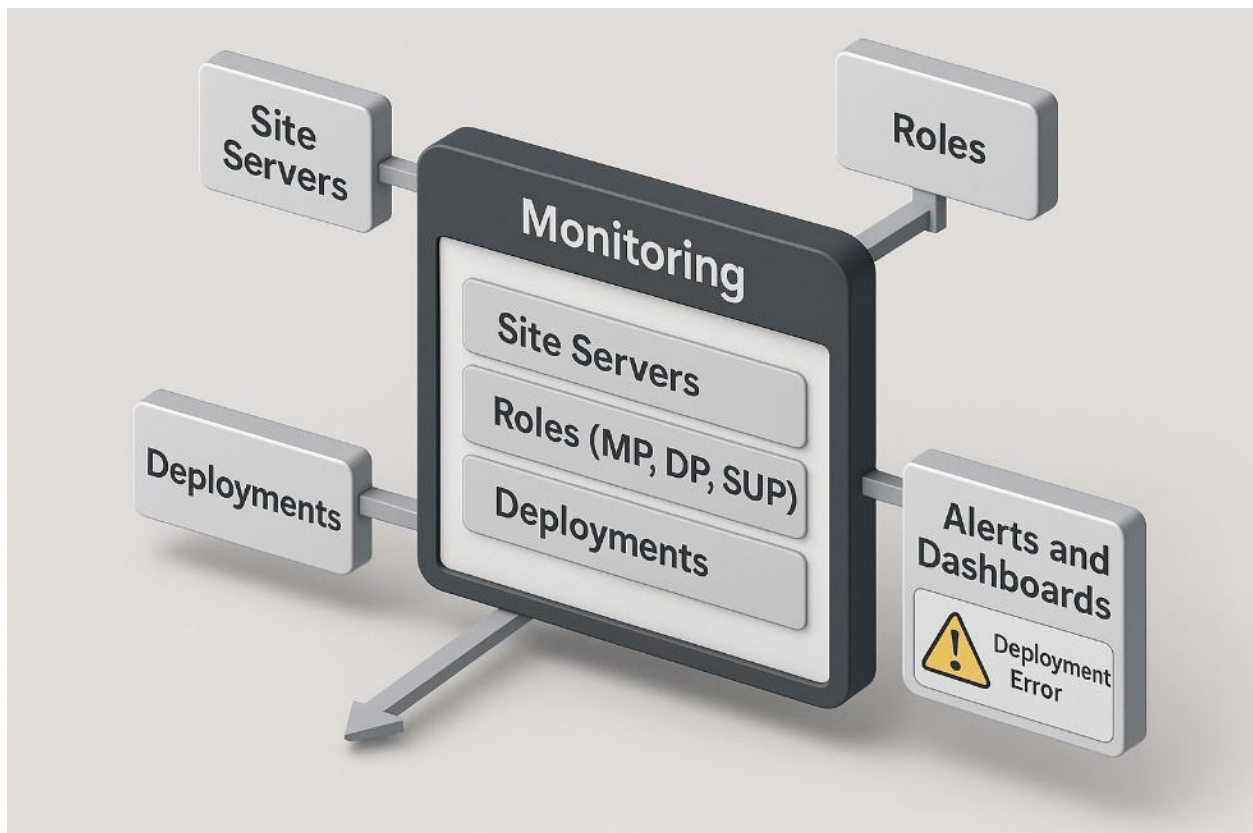
1. **Monitoring** **Workspace** me check karte ho:

- **Site Servers** (running or down)
- **Roles** (MP, DP, SUP status)
- **Deployments** (success/failure)

2. **Dashboards** & **Alerts**

- Site health ka graphical view
- Warnings/Errors directly dikhte hain

3. **Example** → Agar updates fail ho jaate hain → Monitoring me “**Deployment Error**” alert show hota hai.



9. Reviewing Log Files Using Configuration Manager Trace Log Tool

SCCM logs almost everything. To troubleshoot, use **CMTrace** (Log Viewer).

📁 Important Logs:

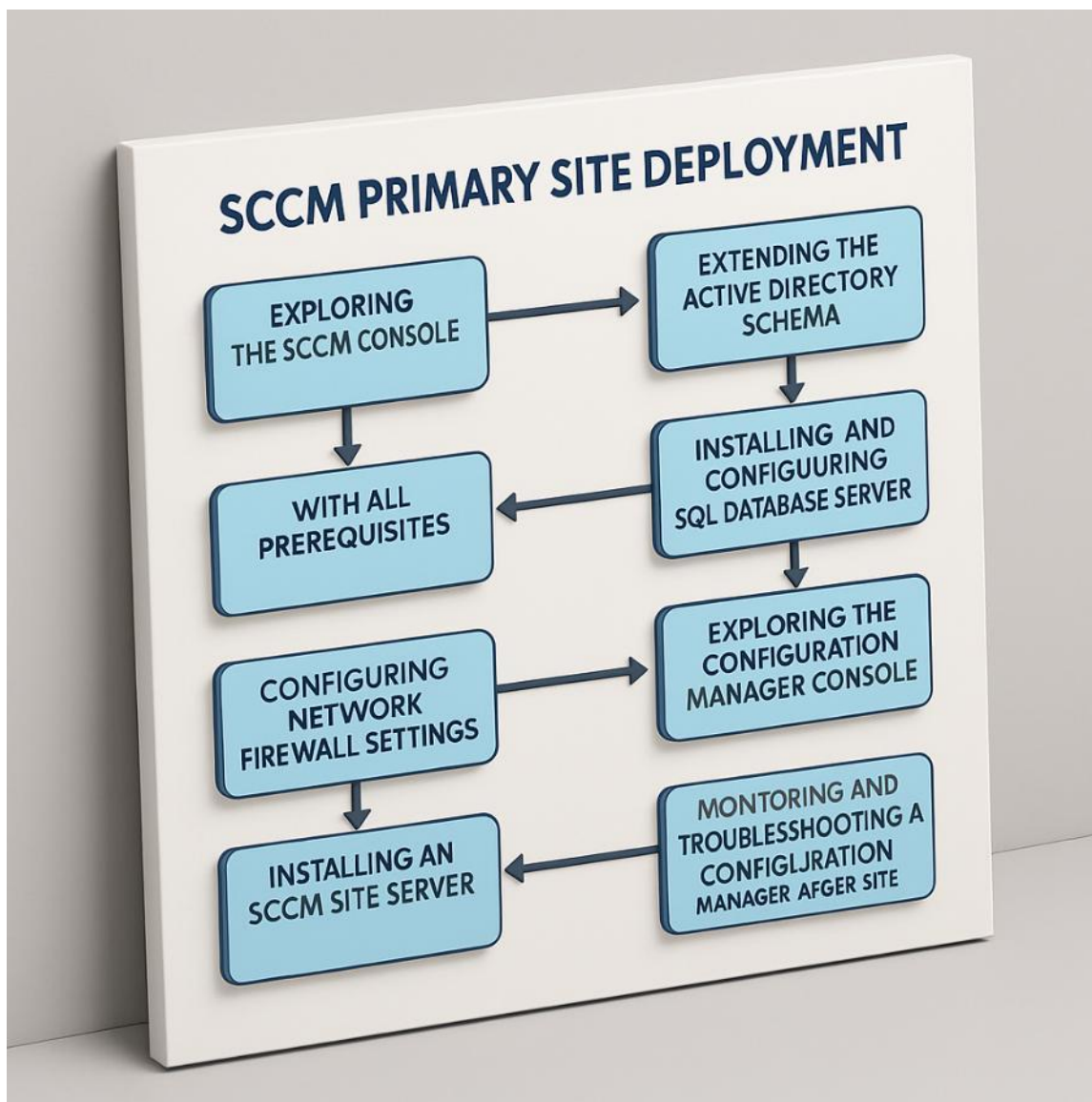
- `ConfigMgrSetup.log` → Installation
- `smsexec.log` → Site server service
- `mpcontrol.log` → Management Point status
- `WUAHandler.log` → Windows Update activity on clients

👉 CMTrace highlights errors in **red**, warnings in **yellow**, so it's easy to analyze.



✓ By the end of Module 2, you'll know how to:

- Prepare environment (AD, SQL, Firewall)
- Install SCCM Primary Site
- Explore Console features
- Troubleshoot using Monitoring + Logs

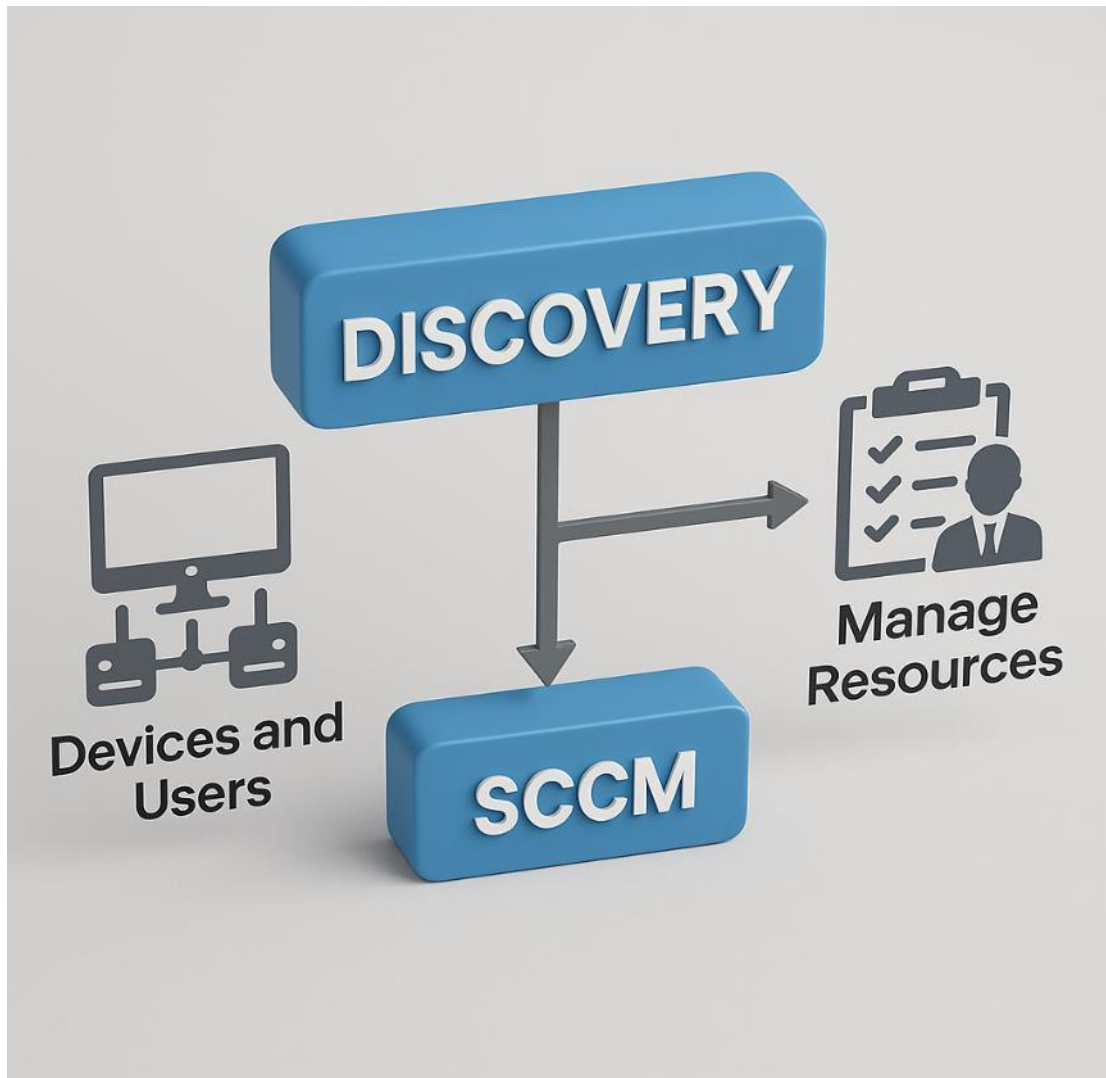


Module 3: Discovering and Organizing Resources

1. Overview of Discovery Methods

Discovery in SCCM = Process of **finding devices and users** in the network so SCCM can manage them.

- Without discovery, SCCM won't know what computers or users exist in your domain.



2. Active Directory Discovery Methods / Network Discovery / Heartbeat Discovery

◇ Active Directory Discovery

- Discovers objects directly from **Active Directory**.
Types:
- **AD Forest Discovery** → Finds all AD sites & subnets.
- **AD Group Discovery** → Finds security groups & members.
- **AD System Discovery** → Finds computers in AD.
- **AD User Discovery** → Finds users in AD.

◇ Network Discovery

- Scans the network to find devices (workstations, routers, printers) even if not in AD.

◇ Heartbeat Discovery

- A “check-in” from SCCM clients → lets SCCM know they are still active.

👉 Example: Every computer sends a **heartbeat** to confirm it's alive.



3. Configuring Discovery Methods

- In SCCM Console → **Administration** → **Discovery Methods**
- Configure what you want to discover:
 - **Forest Discovery** → Finds subnets
 - **Group Discovery** → Finds groups (like "HR Computers")
 - **System Discovery** → Finds machines
 - **User Discovery** → Finds domain users



4. Configuring Boundaries and Boundary Groups in SCCM

- **Boundary** = Defines a **network location** where devices exist.
(Examples: **IP Subnet**, Active Directory Site, IP Address Range, VPN).
- **Boundary Group** = A **collection of boundaries**.
👉 Used to assign clients to the **nearest Distribution Point (DP)** for software/updates.

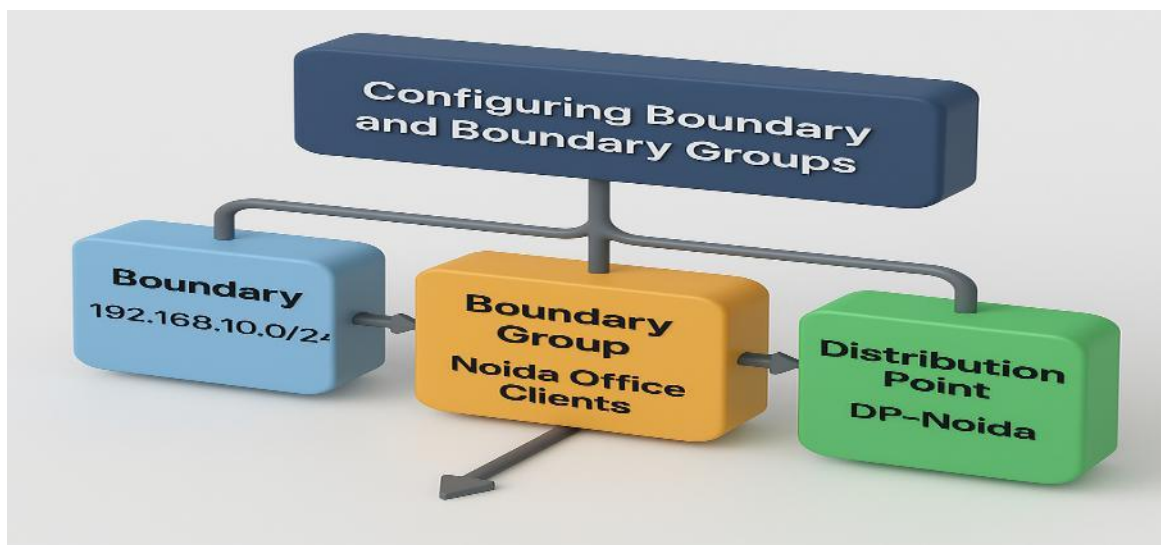
◇ Example:

- **Boundary** → 192.168.1.0/24 (Delhi subnet).
- **Boundary Group** → Add this + other subnets → *Delhi Office Clients*.

- Assigned to → **Distribution Point (DP-Delhi)**

⚡ Production Name (Realistic Example):

- **Boundary:** 192.168.10.0/24 → "Noida Office Subnet"
- **Boundary Group:** "Noida Office Clients" → Assigned to **DP-Noida**



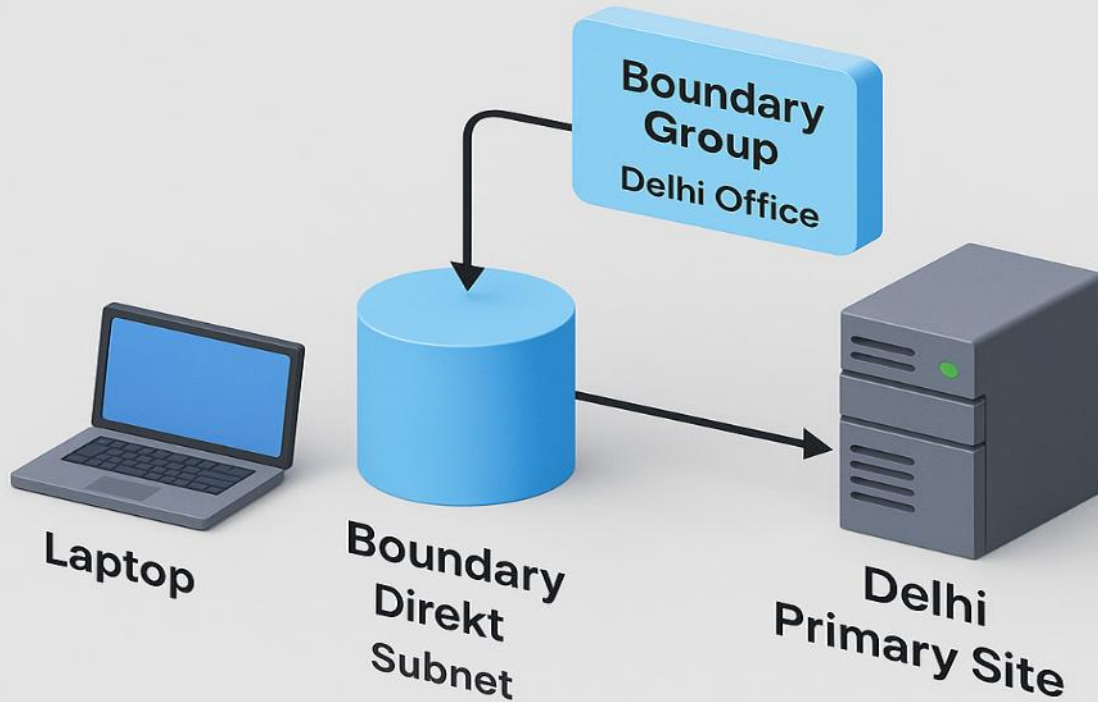
5. Configuring Client Assignments

- Once SCCM discovers devices, they must be assigned to a **Primary Site**.
- Client Assignment ensures the device knows **which SCCM server** to talk to.

👉 Example:

- Laptop in Delhi boundary → Assigned to Delhi Primary Site.

Configuring Client Assignments



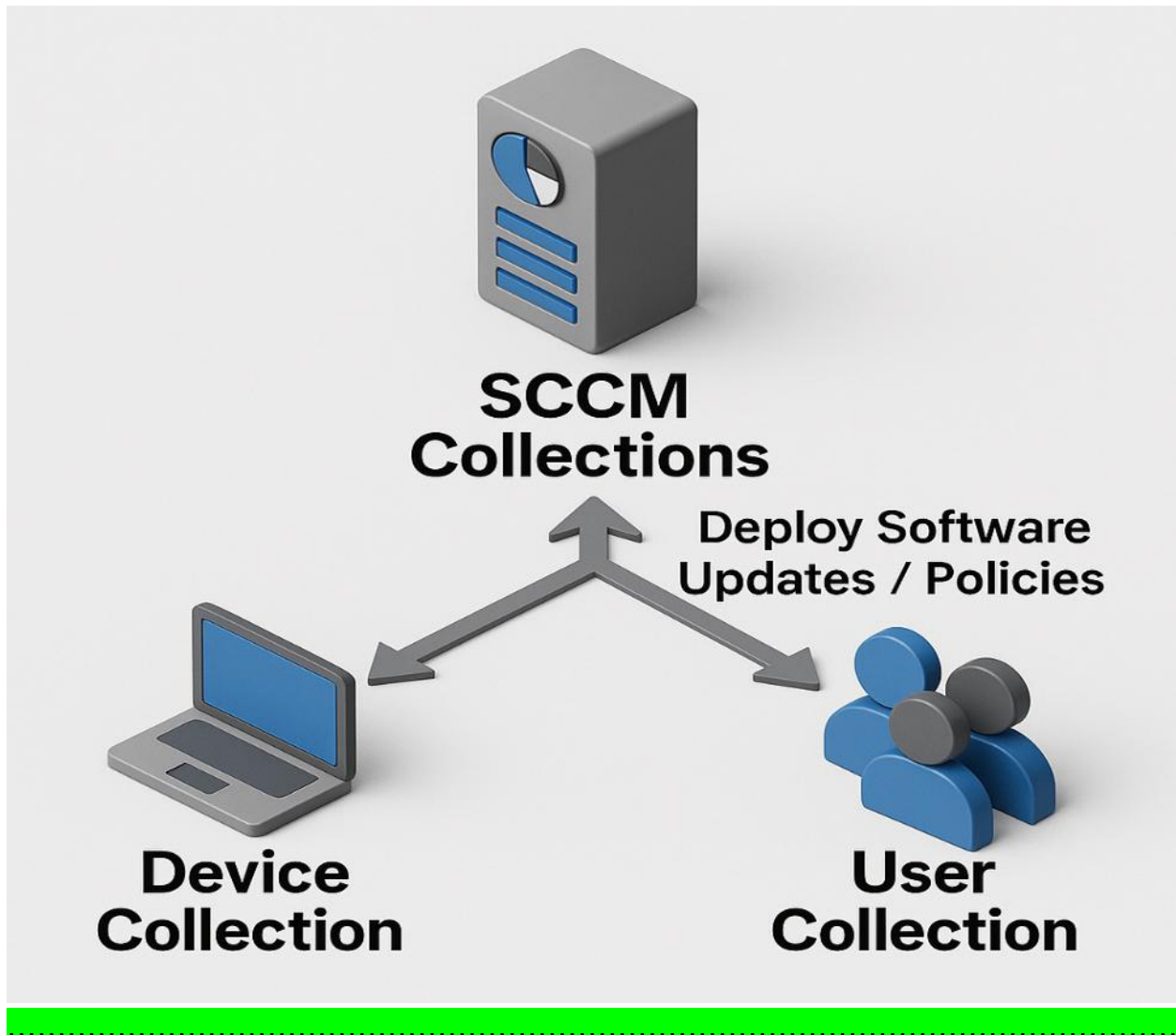
6. Overview and Managing User and Device Collections

- Collections = Groups of devices or users in SCCM.
- Used for targeting software, updates, or policies.

Types:

- **Device Collection** → Group of computers (e.g., “All Windows 11 PCs”).

- **User Collection** → Group of users (e.g., “HR Department Users”).



7. Create Device and User Collection with Membership Rules

Two ways to create collections:

- ◆ Direct Membership Rule

Manually add devices/users one by one.

👉 Example: Add only 5 laptops of managers.

- ◆ Query-based Membership Rule

Automatically includes devices/users that match conditions.

👉 Example: Create query → “All devices with OS = Windows 10” → SCCM auto-updates the collection.

☑ Summary of Module 3:

Discovery finds resources (users, devices, groups).

Boundaries & Groups = Assign devices to nearest site.

Client Assignment = Devices talk to the right SCCM site.

Collections = Organize users/devices for software deployment.

Please give me a 3D diagram, icon-based.



Module 4: Managing the Configuration Manager Client

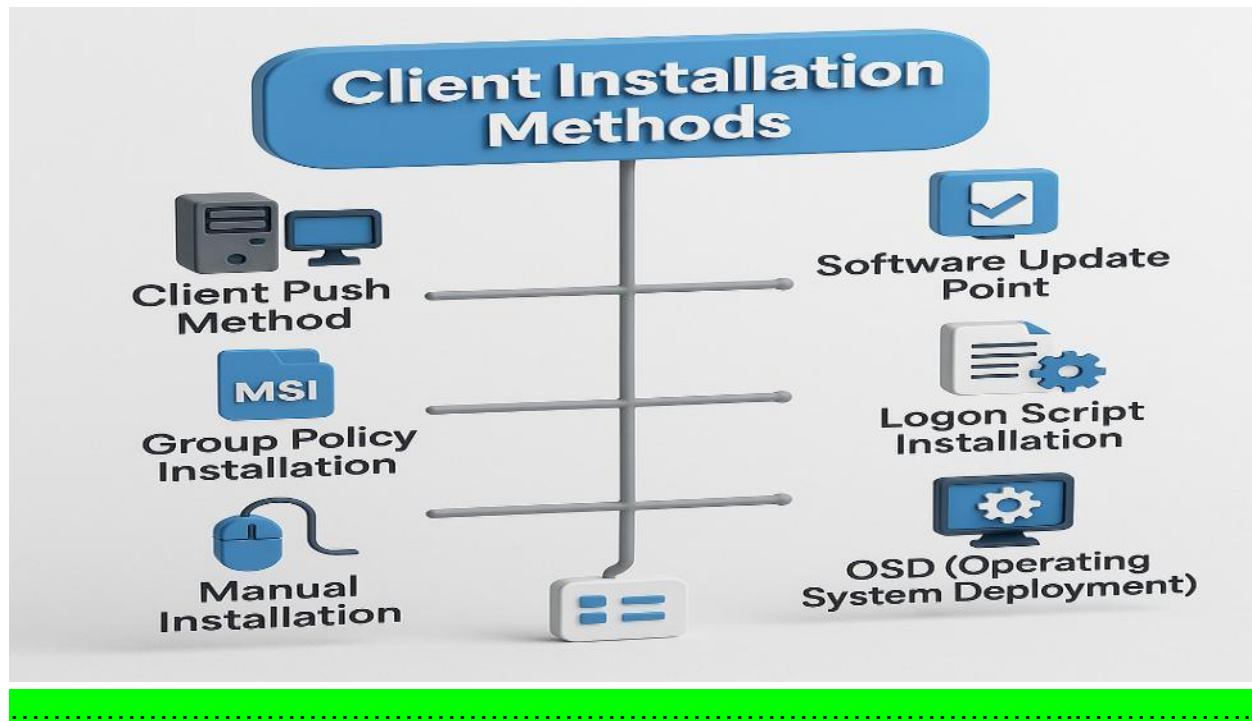
1. Overview of SCCM Client Installation Methods

The SCCM **client agent** is software that gets installed on devices so SCCM can manage them (updates, apps, policies).

◇ Different Methods of Installation:

1. **Client Push Method**
 - SCCM server pushes the client software to devices automatically.
 - Requires admin rights and firewall open.
2. **Site-Wide Push**
 - Automatically pushes clients to all discovered devices in the site.
3. **Software Update Point (SUP) Installation**
 - Client installed via **Windows Update** mechanism.
4. **Group Policy Installation**
 - Deploy MSI package via Active Directory Group Policy.
5. **Manual Installation**
 - Run `ccmsetup.exe` manually on each device.
6. **Logon Script Installation**
 - Script runs when users log in → installs client automatically.
7. **OSD (Operating System Deployment)**
 - Installs client as part of new OS installation.

👉 **Best practice:** Use **Client Push** or **OSD** for automation.



2. Preparing the Site for Client Installation

Before client deployment, ensure:

- Firewall ports are open (SMB 445, RPC 135, HTTP/HTTPS).

Client installation files available at:

`\\SCCMServer\SMS_<SiteCode>\Client`

- Proper permissions (admin rights on target machines).

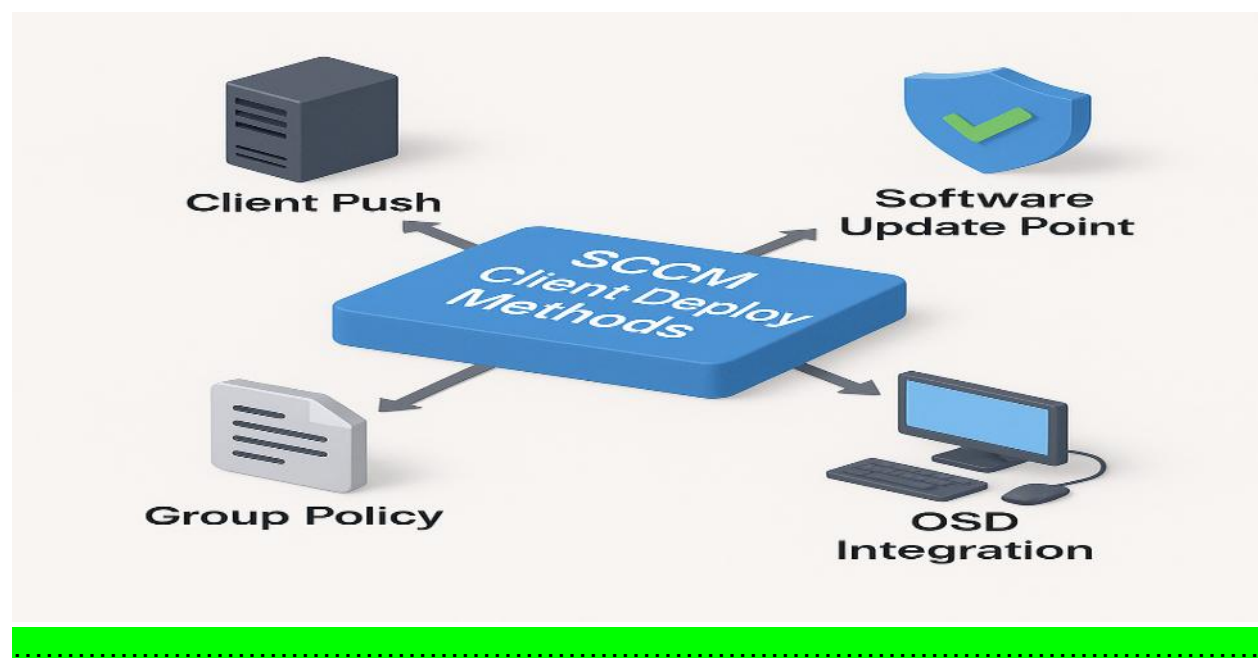


3. Configuring SCCM Client Deployment Methods

Path in SCCM Console:
Administration → Site Configuration → Sites → Client Installation Settings

◇ **Methods to Enable:**

1. **Client Push** → SCCM server pushes client agent to devices.
2. **Software Update Point (SUP)** → Client installs via Windows Update.
3. **Group Policy** → Deploy MSI package through AD Group Policy.
4. **OSD Integration** → Client installs automatically during OS Deployment.



4. Configuring User and Device Policies for SCCM Client Deployment

- Once installed, clients get policies from SCCM.
- Policies define:
 - Software Deployment rules

- Compliance settings
- Endpoint protection rules
- Updates

👉 Example: "All HR Laptops" collection → Apply policy to auto-install Office 365.



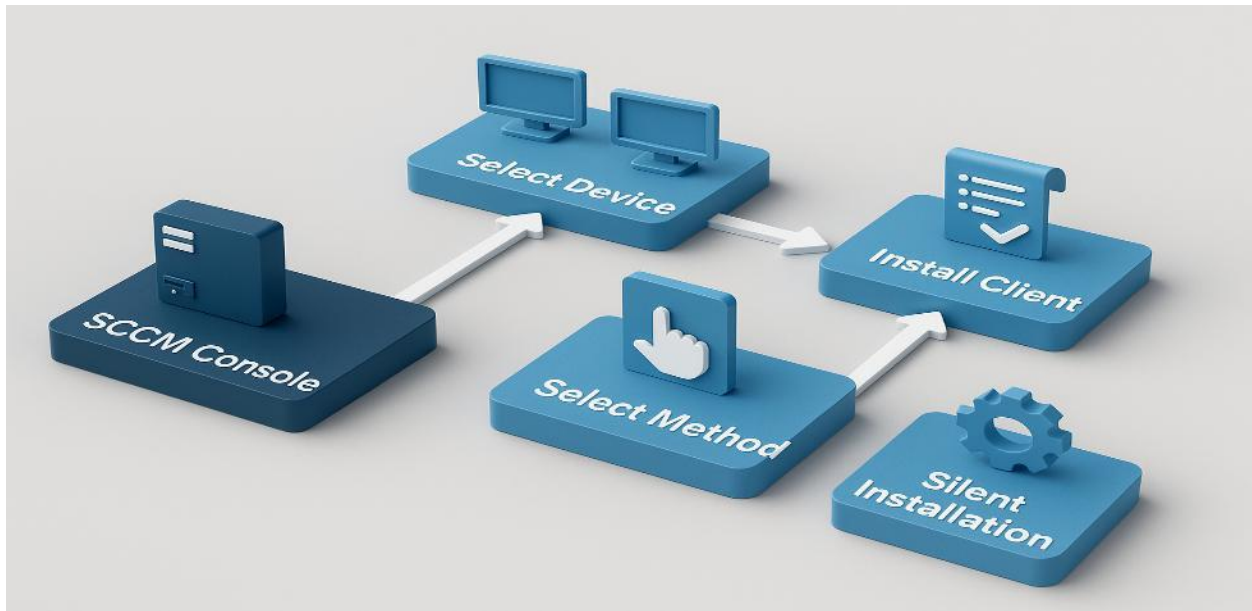
5. Deploying Configuration Manager Clients

◇ Steps:

1. Select Devices in SCCM Console.
2. Choose "Install Client" option.
3. Select Installation Method (Push, Script, Group Policy, OSD, etc.).
4. SCCM installs client silently in the background.

◇ Example:

- Admin selects HR Laptops collection.
- Installs client using Push Method.
- Clients install silently, no user action needed.



6. Configuring and Monitoring Client Installation Process and Status

◇ After Deployment → Monitor Clients

1. **SCCM Console → Monitoring Workspace → Client Status**

- Shows which devices have SCCM client installed and active.

2. **Check Logs on Client Machine:**

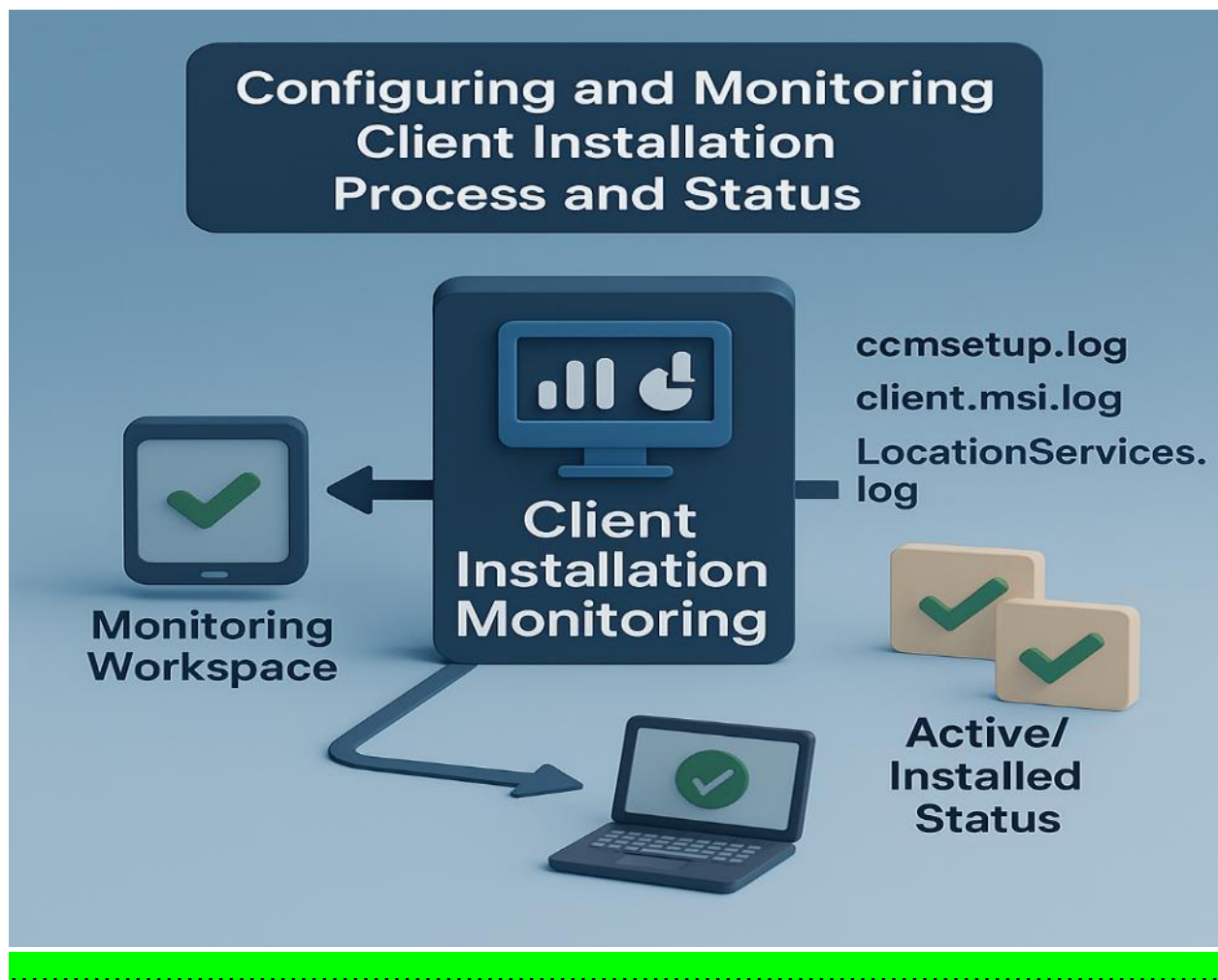
- **ccmsetup.log** → Client installation process.
- **client.msi.log** → MSI installation details.
- **LocationServices.log** → Site assignment check.

3. **Result:**

- Clients show as **Active / Installed** in SCCM Console when successful.

☑ Summary of Module 4

- Different client install methods (**Push, GPO, Manual, OSD**).
- Site preparation (**firewall, files, permissions**).
- Configured deployment methods.
- Applied **user/device** policies.
- Monitoring using **Console + Logs**.



Module 5: Managing Inventory and Software Metering

1. Configuring Asset Intelligence

What is Asset Intelligence?

- A feature in SCCM that provides **extra reports + data** about software and hardware.
- Helps IT admins make decisions about **licenses, usage, and compliance**.

◇ What It Helps You Understand:

1. **Software Licenses** – Track licensed vs installed apps.
2. **Installed Application Versions** – See which app versions exist (e.g., Office 2016, 2019, 2021).
3. **Usage Statistics** – Monitor how often apps are used.

◇ Requirement:

- Enable **Asset Intelligence Synchronization Point** role in SCCM.

◇ Example:

👉 Find out how many devices have **Office 2019 installed** vs how many are actually **licensed**.



2. Configuring Hardware Inventory

- Hardware Inventory = Collects hardware details from client PCs.
- Examples: Processor, RAM, Hard Disk, BIOS, OS version, Serial number.
- Configured in SCCM **Client Settings** → Enable Hardware Inventory → Set schedule (default = every 7 days).

👉 Example: Create a report → “All PCs with less than 8GB RAM.”



3. Configuring Software Inventory

◇ What is Software Inventory?

- Scans client PCs for installed applications and files.

◇ Data Collected:

- Executable files (.exe)
- Version details
- File path/location

◇ Configuration Path:

- In SCCM Client Settings → Enable Software Inventory

◇ Example Use Case:

👉 Generate a report: “All machines with Chrome installed.”



4. Configure and Manage Hardware and Software Inventory, Monitor Inventory Data Flow

◇ Inventory Data Flow

1. **Client PC** → Runs **inventory cycle** (collects hardware + software data).
2. **Data sent** → **To Management Point (MP)**.
3. **MP sends** → **To Primary Site Database (SQL)**.
4. **Reports generated** → **In SCCM Console**.

◇ Monitoring Inventory

- **On Client Logs:**
 - `InventoryAgent.log` → Tracks hardware/software inventory cycles.
- **In Console:**
 - Path: *Monitoring* → *Reporting* → *Inventory Reports*.

◇ Example Use Case

👉 Admin checks a report: “All PCs with less than 8GB RAM and Chrome installed.”



5. Software Metering – Overview, Benefits & Configuration

◇ Overview

- **Software Metering** = Tracks **how often** and **how long** applications are used on client PCs.

◇ Benefits

1. **Identify unused licenses** → Helps reduce cost.
2. **Track software compliance** → Ensure proper usage of licensed software.

◇ Configuration

- In **Client Settings** → **Software Metering** → **Enable**
- Define **specific rules** → e.g., Track **Photoshop.exe** usage.

◇ Example

👉 If only **20 out of 100 users** are actively using Photoshop → Company can **reduce license purchases**.



6. Lab: Configuring and Managing Inventory Collection

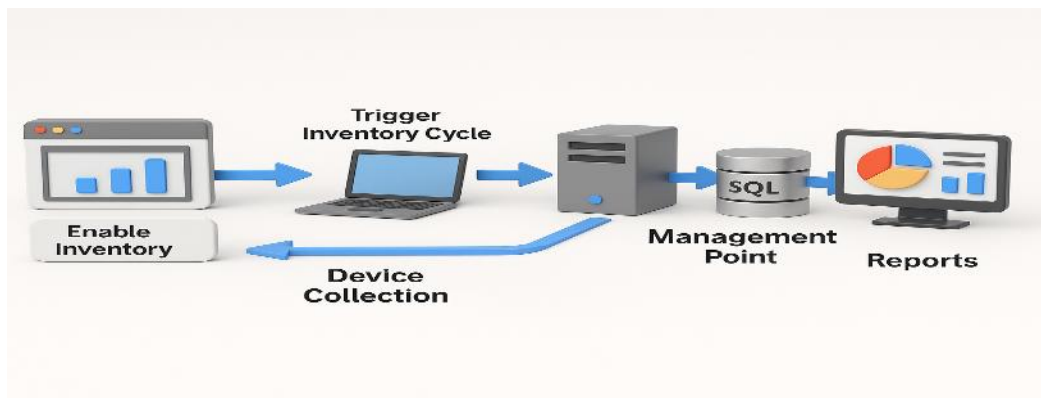
◇ Practical Steps:

1. Go to **Administration** → **Client Settings**.
2. Enable **Hardware Inventory** & **Software Inventory**.
3. Deploy client settings to a **Device Collection**.
4. On client machine → Trigger **Inventory Cycle**:
 - Control Panel → Configuration Manager → **Actions Tab**.
5. Verify results in **SCCM Reports**.

☒ Summary of Module 5

- **Asset Intelligence** → Software licensing & usage reports.

- **Hardware Inventory** → Collects hardware details (CPU, RAM, Disk).
- **Software Inventory** → Collects installed software info.
- **Inventory Data Flow** → Client → MP → SQL DB → Reports.
- **Software Metering** → Tracks actual usage of apps (license optimization).



Module 6: Managing Software Deployments (Application & Package Model).

◇ 1. Introduction to Applications and Packages Models

- **Application Model (new)** → Used for modern apps, supports detection method, dependencies, approval flow.
- **Package Model (legacy)** → Simple, used for older scripts/programs.

👉 Example:

- **Application Model** → Deploy *Office 365*.
- **Package Model** → Deploy *custom script* or *WinRAR*.

◇ 2. Configuring Software Distribution / Concepts

- Software Distribution = Process of delivering apps, updates, or programs to client devices via SCCM.
- Uses **Distribution Points (DPs)** to deliver content.

◇ 3. Creating & Deploying Application with Approval Flow

- Create app in SCCM Console.
- Define **Detection Method** (check if app is installed).
- Define **Requirements** (e.g., OS must be Windows 11, 8GB RAM).
- Configure **Approval Flow** (users request app, admin approves).



Example:

HR user requests *Zoom App* → Admin approves → Zoom installs automatically.

◇ 4. Distributing Content to Distribution Points

- After creating an app, send its content to **DPs**.
- Clients download app from nearest DP.



Example:

Delhi users get app from **DP-Delhi**, Noida users from **DP-Noida**.

◇ 5. Deploying an Application to a User or a Device

- Target a **User Collection** or **Device Collection**.
- Example: Deploy *Office 365* → HR Department Users.

◇ 6. What is Software Center?

- Client-side app where users see available applications, updates, and installations.

👉 Example: User opens **Software Center** → Clicks *Install Chrome*.

◇ 7. Configuring the Application Catalog

- Used to allow users to browse/request software.
- Works with approval flow.

◇ 8. Configuring Packages and Programs

- Create legacy **packages/programs** for simple deployments.
- Example: Deploy *script* or *WinRAR installer*.

◇ 9. Managing Application Deployment

- Check deployment status in **Monitoring Workspace**.
- See which devices/users succeeded or failed.

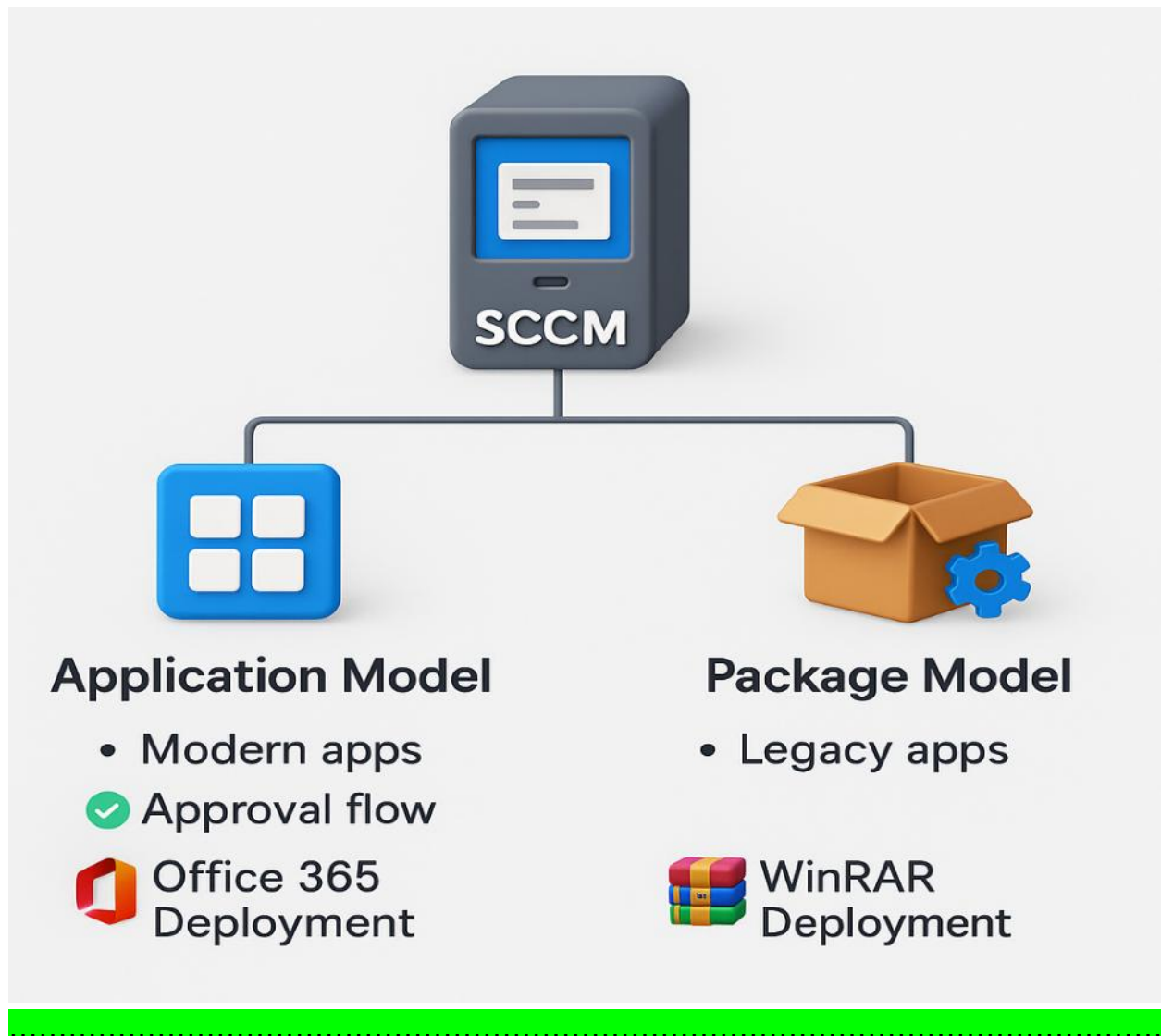
◇ 10. Viewing Log Files & Troubleshooting

- Important logs:
 - **AppDiscovery.log** → Detection method results.
 - **AppEnforce.log** → Application installation details.
 - **DistMgr.log** → Distribution point logs.



Summary:

Module 6 teaches how to **create, distribute, deploy, and monitor apps/packages** using SCCM.

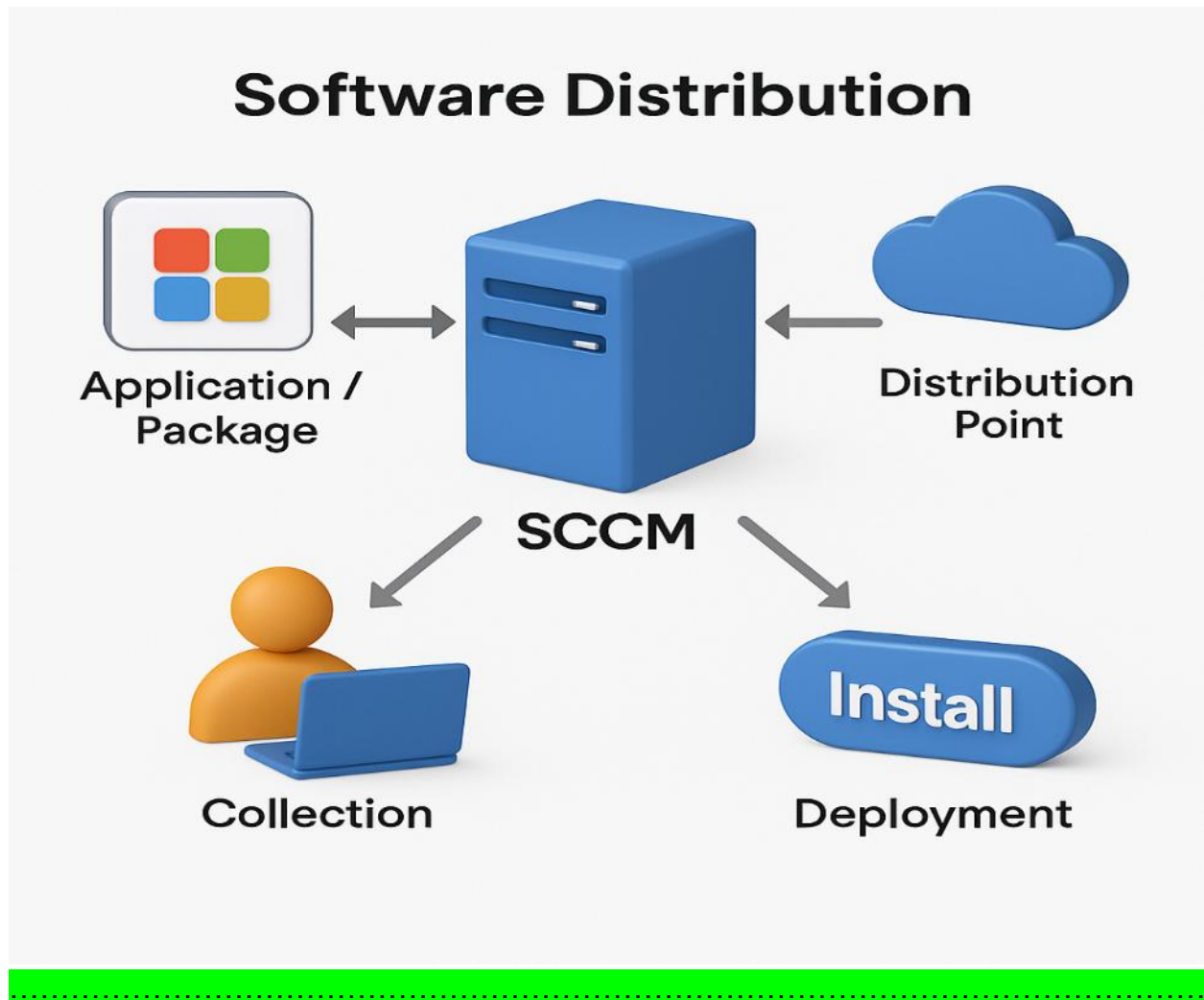


2. Configuring Software Distribution / Software Distribution Concepts

- **Software Distribution** = Delivering apps/packages from SCCM → clients.
- Components:
 1. **Application / Package** → What to deploy
 2. **Distribution Point (DP)** → Where content is stored

3. **Collection** → Who gets it (users/devices)

4. **Deployment** → Action (install/uninstall)



3. Creating and Deploying Application with Approval Flow, Detection, and Requirements

- **Approval Flow** → Some apps require user/manager approval before install.
- **Detection Method** → Ensures app installs only if not already present.
👉 Example: Detection rule checks for `winword.exe` for MS Office.
- **Deployment Requirement** → Conditions before install.
👉 Example: Install only on Windows 10 with ≥ 8GB RAM.



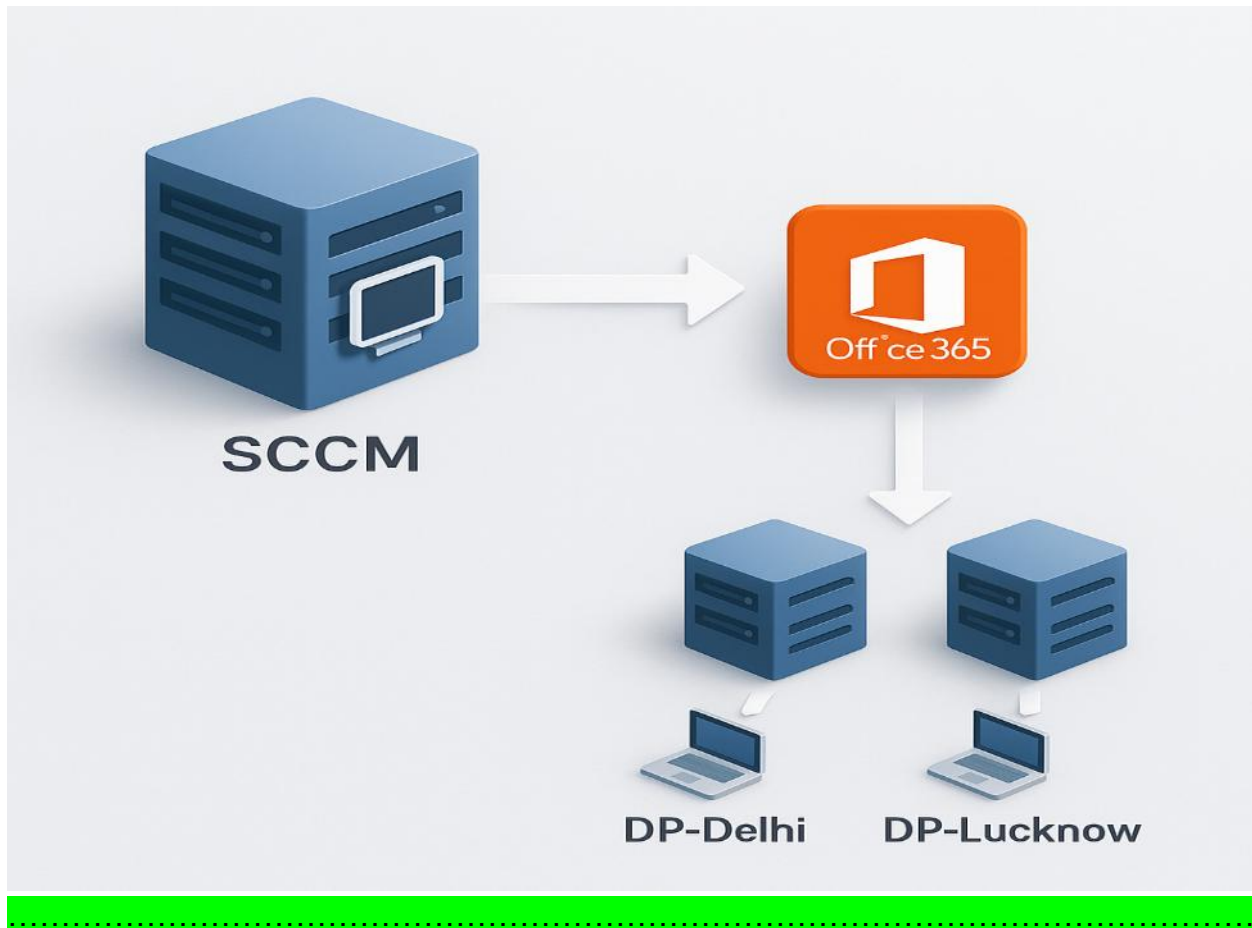
4. Distributing Content to Distribution Points

◇ Concept

- Before you **deploy software**, you must **send the application/package** to **Distribution Points (DPs)**.
- **DPs = Servers** where clients download the software from.

◇ **Example**

- Application: **Office 365**
- Distributed to: **DP-Delhi** and **DP-Lucknow**
- Clients in Delhi and Lucknow will download from their **nearest DP**.



5. Deploying an Application to a User or Device

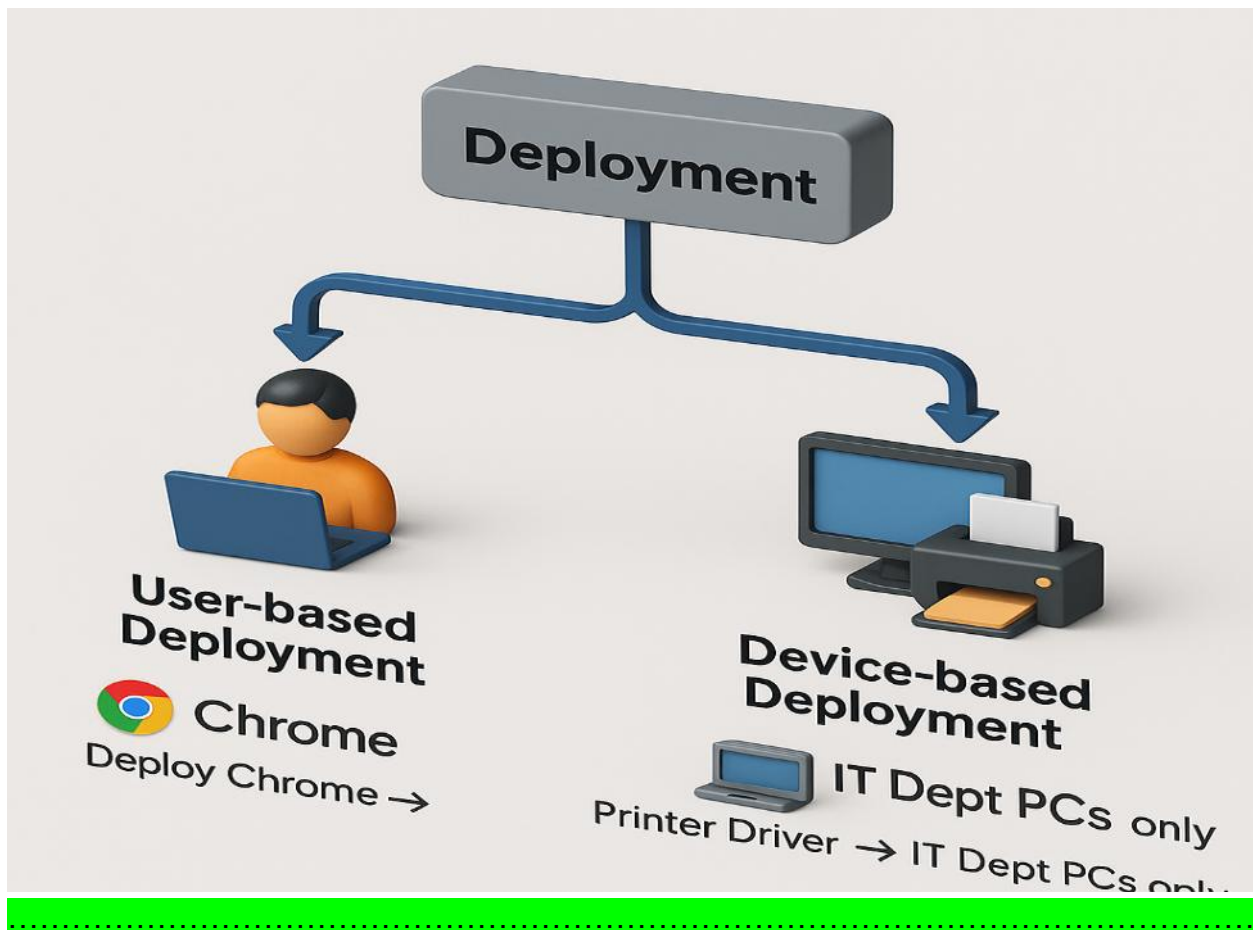
◇ **User-based Deployment**

- Software **follows the user.**
- Installed wherever the user logs in.

- 👉 Example: Deploy **Chrome** → *All Users.*

◇ Device-based Deployment

- Software installed on **specific PCs only.**
- Stays with the machine, not the user.
- 👉 Example: Deploy **Printer Driver** → *IT Dept PCs only.*



6. What is a Software Center?

Definition:

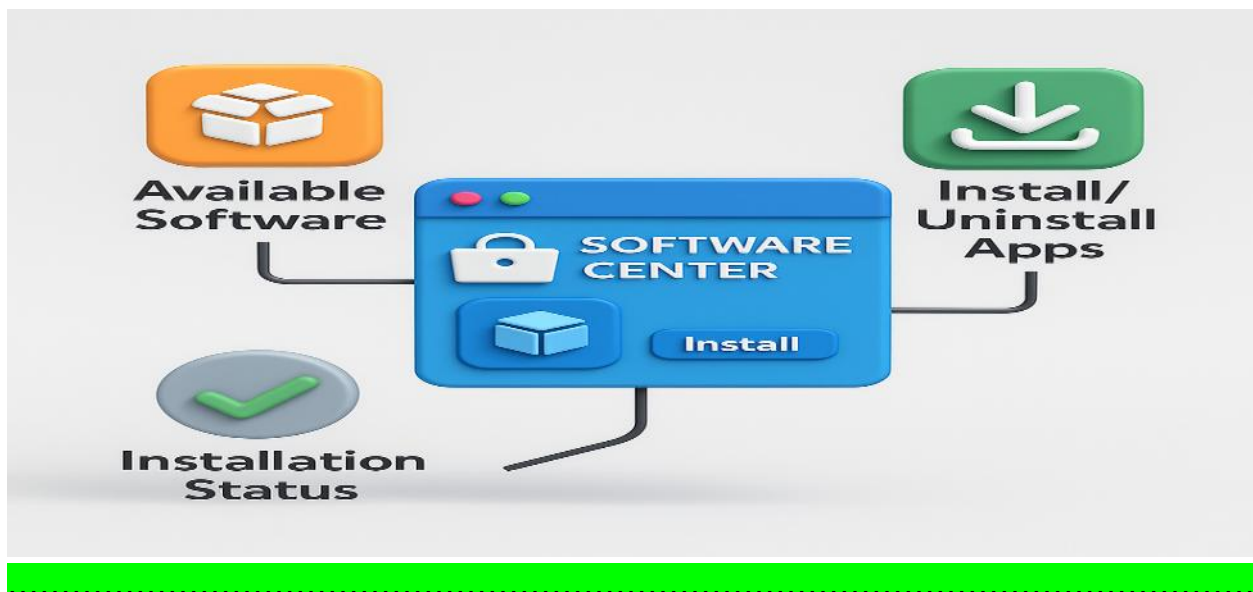
- **Software Center** is a **client-side application** installed with the SCCM client.
- It acts like a **self-service app store** for employees.

◇ **What Users Can Do in Software Center:**

1. **See Available Software** → List of apps published by IT.
2. **Install / Uninstall Applications** → Directly from Software Center.
3. **View Installation Status** → Check if app is installing, failed, or completed.

◇ **Example**

👉 Employee opens **Software Center** → Installs **Zoom** → Status shows *Installing* → *Installed*.



7. Configuring the Application Catalog

◇ **What is Application Catalog?**

- A **web portal** where users can **browse, request, and install apps**.
- Works with **Approval workflows** → Managers/IT can approve before install.

◇ **Key Features:**

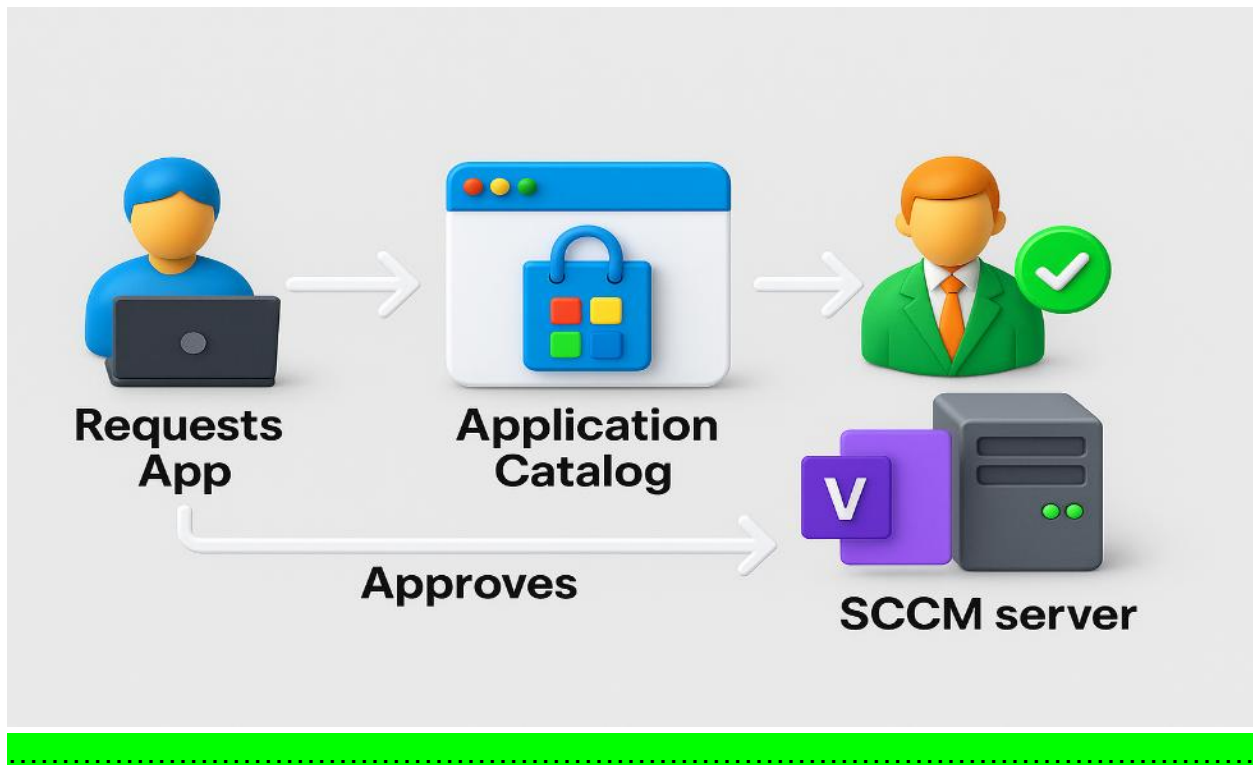
1. Users browse available applications.

2. Request an app → Approval workflow.

3. Once approved → SCCM installs automatically.

◇ Example

👉 Employee requests **Visio** from the portal → Manager approves → SCCM installs Visio on employee's laptop.



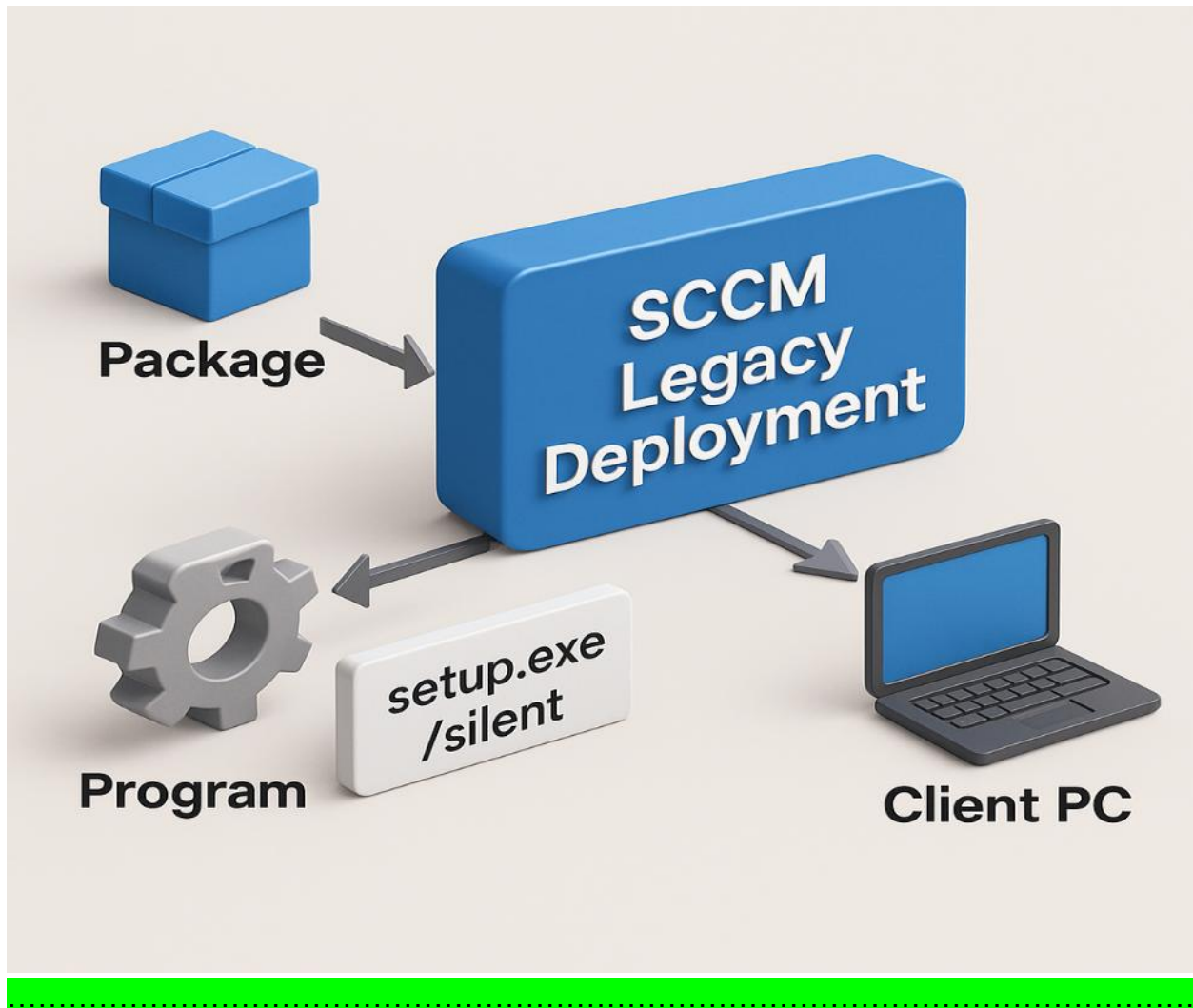
8. Configuring Packages and Programs

◇ What is it?

- Used for **older/legacy deployments** (before modern Application Model).
- **Package** = Container for the software files.
- **Program** = Command line instructions for how to install/uninstall the package.

◇ Example

- **Package:** *7zip* *installer* *files*
- **Program:** `setup.exe /silent` (runs the installation silently).



9. Managing Application Deployment

◇ After Deployment Steps:

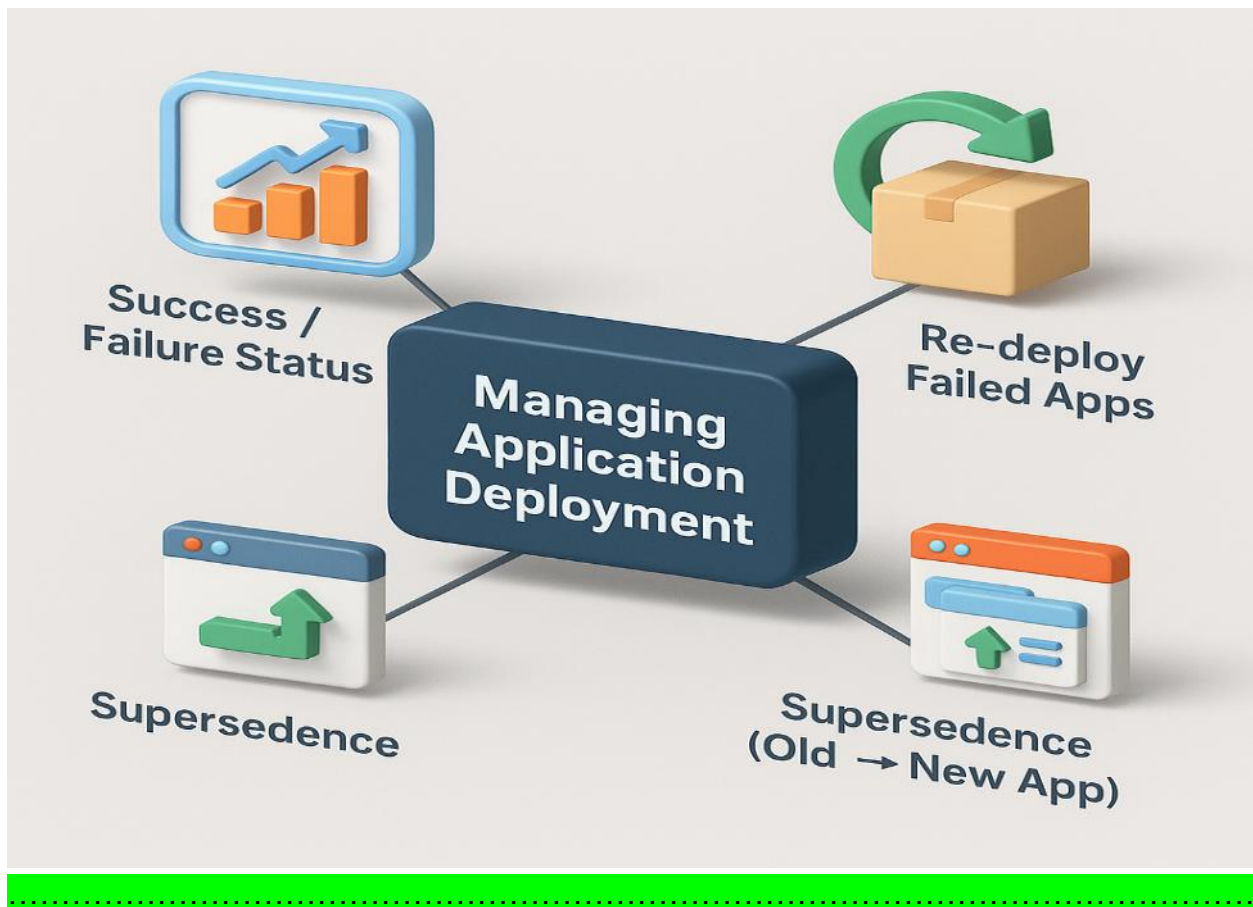
1. **Monitor** **Deployment** **Status**
 - Go to **Monitoring Workspace** → Check success vs failure.

2. Re-deploy **if** **Failed**

- If some clients failed → Admin can re-deploy the application.

3. Update Applications (Supersedence)

- New version replaces old version automatically.
- Example: *Office 2016* → *Office 2019*.



10. Viewing Log Files, Monitoring and Troubleshooting Application Deployment

- ◇ Why Logs?

- Logs are the **main way to troubleshoot failed application deployments** in SCCM.
- They tell you **where and why** the failure happened (client-side or server-side).

Important Client-Side Logs

- **AppDiscovery.log** → Shows **detection method results**.
- **AppEnforce.log** → Records **install/uninstall actions** of applications.
- **CAS.log** → Tracks **content download** progress.
- **ExecMgr.log** → Handles **program execution details**.

Important Site Server Logs

- **distmgr.log** → Shows **content distribution to Distribution Points (DPs)**.
- **smsexec.log** → Tracks **overall application processing**.

Summary of Module 6

1. **Applications (modern)** vs **Packages (legacy)**
2. **Distribution flow** → **DPs** → **Collections** → **Deployment**
3. **Detection & Requirement rules** → **Ensure accuracy**
4. **Software Center & Application Catalog** → **User-facing tools**
5. **Logs + Monitoring** → **Critical for troubleshooting**

Viewing Log Files, Monitoring and Troubleshooting Application Defects

